

MATH 74: ALGEBRAIC TOPOLOGY

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1. TOPOLOGICAL SPACES & HOMEOMORPHISMS

Definition 1.1. A **topology** on a set X is a collection J of subsets of X having the following properties.

- (1) $\emptyset, X \in J$
- (2) The union of the elements of any sub-collection of J is in J .
- (3) The intersection of finitely many elements in J is in J .

A set for which a topology has been specified is said to be a topological space. We think of it as a pair (X, J) .

Definition 1.2. Elements of J are called **open sets** of the topological space (X, J) .

Definition 1.3. A subspace Y of a topological space (X, J) is a subset $Y \subset X$ which is given the **subspace topology** where the open sets of $Y = \{U \cap Y : U \in J\}$

Definition 1.4. A space X is **path connected** if every pair of points of X can be joined by a path in X .

Definition 1.5. Let (X_1, J_1) and (X_2, J_2) be topological spaces. A map $f : (X_1, J_1) \rightarrow (X_2, J_2)$ is said to be **continuous** if $f^{-1}(U) \in J_1$ for all $U \in J_2$.

Definition 1.6. Let (X, J) and (X', J') be topological spaces. Then they are **homeomorphic** if $\exists$ a continuous bijection $f : (X, J) \rightarrow (X', J')$. So that f^{-1} is also continuous.

Example 1.1. Show that any two open intervals of $\mathbb{R}$ are homeomorphic (where we put subspace topology on the open intervals).

Solution. Let (a, b) and (c, d) be open intervals and define $f : (a, b) \rightarrow (c, d)$ as follows:

$$f(x) = c + (x - a) \frac{d - c}{b - a}$$

Then, f is continuous since it is a linear function in x , bijective because it is strictly increasing (injective) and has a continuous inverse.

Remark 1.7. Number of path-connected topologies is preserved by continuous maps. For instance, $[0, 1]$ is not homeomorphic to $(0, 1)$, since $[0, 1] \setminus \{0\} = (0, 1]$ is path-connected but $(0, 1) \setminus \{pt\} = (0, pt) \cup (pt, 1)$ which has two path-connected components. Similarly, $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^2$ since $\mathbb{R} \setminus \{pt\}$ gives us two disconnected components, but $\mathbb{R}^2 \setminus \{pt\}$ is path-connected.

Example 1.2.

- (a) Is $\mathbb{R}^2$ homeomorphic to $\mathbb{R}^3$?
- (b) Is $S^2 (= x^2 + y^2 + z^2 = 1)$ homeomorphic to $T^2 = S^1 \times S^1$?
- (c) Is S^2 homeomorphic to $S^1 \times [a, b]$?

Solution.

- (a) No, since $\pi_1(\mathbb{R}^2 \setminus \{\text{pt}\}) = \pi_1(S^1) = \mathbb{Z}$ and $\pi_1(\mathbb{R}^3 \setminus \{\text{pt}\}) = \pi_1(S^2) = 0$
- (b) No, since $\pi_1(S^2) = 0$ and $\pi_1(T^2) = \mathbb{Z} \oplus \mathbb{Z}$.
- (c) No, since $\pi_1(S^2) = 0$ and $\pi_1(S^1 \times [a, b]) = \pi_1(S^1) = \mathbb{Z}$.

Definition 1.8. If f and f' are continuous maps from spaces X to Y we say that f is homotopic to f' if $\exists$ a continuous map $F : X \times [0, 1] \rightarrow Y$ such that $F(x, 0) = f(x)$ and $F(x, 1) = f'(x)$. F is called the **homotopy** between f and f' , and we write $f \simeq f'$.

Definition 1.9. If X and Y are topological spaces, we can define a **product topology** on $X \times Y$ by saying that a basis consists of the subsets $U \times V$ as U ranges over open sets in X and V ranges over open sets in Y .

Definition 1.10. Define a **path** f as $f : [0, 1] \rightarrow X$ with $f(0) = x_0$ and $f(1) = x$.

Definition 1.11. Two paths $f, f' : [0, 1] \rightarrow X$ are said to be path homotopic if there exist a continuous map $F : I \times I \rightarrow X$ such that $\forall s \in I$,

$$F(s, 0) = f(s), \quad F(s, 1) = f'(s)$$

and $\forall t \in I$,

$$F(0, t) = x_0, \quad F(1, t) = x$$

Then we write $f \underset{p}{\simeq} f'$.

Example 1.3. Let $f : X \rightarrow \mathbb{R}^2$ and $g : X \rightarrow \mathbb{R}^2$ then $f \simeq g$.

Proof. Want to find F such that $F : X \times I \rightarrow \mathbb{R}^2$, $F(x, 0) = f(x)$, and $F(x, 1) = g(x)$ for all $x \in X$. The following map (straight-line homotopy) works.

$$F(x, t) = (1 - t)f(x) + tg(x)$$

□

Lemma 1.12. The relations $\simeq$ and $\simeq_p$ are equivalence relations.

Proof.

Homotopy is an equivalence relation:

- (1) **Reflexive:** $f \simeq f$ by $F(s, t) = f(s)$.
- (2) **Symmetric:** Let $f \simeq g$ by the homotopy $F(s, t)$. Then $g \simeq f$ is homotopic by the homotopy $G(s, t) = F(s, 1 - t)$.
- (3) **Transitive:** Let $f \simeq g$ by $F(s, t)$ and $g \simeq h$ by $G(s, t)$. We can show that $f \simeq h$ by

$$H(s, t) = \begin{cases} F(s, 2t), & 0 \leq t \leq 1/2 \\ G(s, 2t - 1), & 1/2 \leq t \leq 1 \end{cases}$$

Then $H(s, 0) = F(s, 0) = f(s)$, and $H(s, 1) = G(s, 1) = h(s)$. Hence, $f \simeq h$.

Path-Homotopy is an equivalence relation:

- (1) **Reflexive:** $f \simeq_p f$ by $F(s, t) = f(s)$.

- (2) **Symmetric:** Let $f \simeq_p g$ by the homotopy $F(s, t)$. Then $g \simeq_p f$ is homotopic by the homotopy $G(s, t) = F(s, 1 - t)$.
- (3) **Transitive:** Let $f \simeq_p g$ by $F(s, t)$ and $g \simeq_p h$ by $G(s, t)$. We can show that $f \simeq_p h$ by,

$$H(s, t) = \begin{cases} F(s, 2t), & 0 \leq t \leq 1/2 \\ G(s, 2t - 1), & 1/2 \leq t \leq 1 \end{cases}$$

Where

- $H(s, 0) = F(s, 0) = f(s)$
- $H(s, 1) = G(s, 1) = h(s)$
- $H(0, t) = x_0$ and $H(1, t) = x_1$.

Hence, $f \simeq_p h$.

Thus path-homotopies are equivalence relations and we write $[f]$ to denote the path-homotopy equivalence class of f . $\square$

Lemma 1.13 (Pasting Lemma). *Let $X = A \cup B$ be a topological space with $A, B \subset X$ are closed. Let $f : A \rightarrow Y, g : B \rightarrow Y$ be continuous and $f(x) = g(x), \forall x \in A \cap B$, then*

$$h(x) = \begin{cases} f(x), & x \in A \\ g(x), & x \in B \end{cases}$$

is continuous.

Definition 1.14. *Let $f : I \rightarrow X$ be a path from x_0 to x_1 and $g : I \rightarrow X$ be a path from x_1 to x_2 , then we define a new path*

$$f \star g : I \rightarrow X$$

from x_0 to x_2 by

$$f \star g(t) = \begin{cases} f(2t), & t \in [0, 1/2] \\ g(2t - 1), & t \in [1/2, 1] \end{cases}$$

*where $\star$ is **composition of paths**.*

Some Properties of ' $\star$ '.

- (1) ' $\star$ ' is well-defined operation in the path homotopy equivalence class,

$$[f] \star [g] = [f \star g]$$

- (2) Identity:

$$[f] \star [e_{x_1}] = [f], \quad [e_{x_0}] \star [f] = [f]$$

where e_{x_t} is the constant path based at $x \in X$ and $t \in [0, 1]$

- (3) Inverse:

$$[f] \star [\bar{f}] = [e_{x_0}], \quad [\bar{f}] \star [f] = [e_{x_1}]$$

where $\bar{f}(t) = f(1 - t)$.

- (4) Associative:

$$[f] \star ([g] \star [h]) = ([f] \star [g]) \star [h]$$

Proof.

- (1) Let $f \simeq_p f'$ by F and $g \simeq_p g'$ by G , then we can show that $f \star g \simeq_p f' \star g'$ by defining $H : I \times I \rightarrow X$ such that

$$H(s, t) = \begin{cases} F(2s, t), & s \in [0, 1/2] \\ G(2s - 1, t), & s \in [1/2, 1] \end{cases}$$

- (2) $f \star e_{x_1} \simeq_p f$ by $H : I \times I \rightarrow X$ such that

$$H(s, t) = \begin{cases} f\left(\frac{2s}{2-t}\right), & s \in [0, \frac{2-t}{2}] \\ x_1, & s \in [\frac{2-t}{2}, 1] \end{cases}$$

- (3) To see $e_{x_0} \simeq_p f \star \bar{f}$ define a path $\alpha(t)$ that starts at x_0 and ends at $f(t)$. Then the path-homotopy is given by $\alpha(t) \star \bar{\alpha}(t)$.

$$H(s, t) = \begin{cases} f(2st), & s \in [0, 1/2] \\ f(2t(1-s)), & s \in [1/2, 1] \end{cases}$$

- (4) By well-definedness,

$$[f] \star ([g] \star [h]) = [f] \star [g \star h] = [f \star g \star h] = [f \star g] \star h = ([f] \star [g]) \star [h]$$

□

Remark 1.15. Note that $\star$ operation on the set of path-homotopy equivalence classes of paths in the space X forms a groupoid structure.

2. THE FUNDAMENTAL GROUP

Definition 2.1. Let X be a space and $x_0 \in X$ be a point in X we say that a path $f : I \rightarrow X$ is a **loop** based at x_0 if $f(0) = f(1) = x_0$

Definition 2.2. The set of all path-homotopy equivalence classes of loops based at x_0 forms a group under $\star$ called the **fundamental group** of x based at x_0 , denoted as $\pi_1(X, x_0)$

- Identity: $[e_{x_0}]$
- Inverses: $[f] \rightarrow [\bar{f}]$

Example 2.1. $\pi_1(\mathbb{R}^n, (0, 0)) \equiv 0 \equiv \langle e_{origin} \rangle$

Proof. Take any two loops $f(t), g(t)$ based at the origin. Then consider the straight-line homotopy,

$$H(s, t) = f(s)(1 - t) + tg(s)$$

where,

- $H(s, 0) = f(s)$
- $H(s, 1) = g(s)$
- $H(0, t) = f(0)(1 - t) + tg(0) = (0, 0) = H(1, t)$

Hence, it is a path-homotopy. □

Definition 2.3. Let α be a path from x_0 to x_1 , where $x_0, x_1 \in X$. We define a map $\hat{\alpha} : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ such that

$$\hat{\alpha}([f]) := [\bar{\alpha}] \star [f] \star [\alpha] = [\bar{\alpha} \star f \star \alpha]$$

Proposition 2.4. $\hat{\alpha}$ is well-defined and an isomorphism.

Proof. Let $f \simeq_p f'$. We need to check that

$$\begin{aligned} [\bar{\alpha}] \star [f] \star [\alpha] &= [\bar{\alpha}] \star [f'] \star [\alpha] \\ \iff [\bar{\alpha} \star f \star \alpha] &= [\bar{\alpha} \star f' \star \alpha] \\ \iff \bar{\alpha} \star f \star \alpha &\simeq_p \bar{\alpha} \star f' \star \alpha \end{aligned}$$

Since $\star$ is well-defined, $[f] \star [\alpha] = [f'] \star [\alpha]$, $\hat{\alpha}$ is well-defined

Let $[f], [g] \in \pi_1(X, x_0)$.

$$\begin{aligned} \hat{\alpha}([f]) \star \hat{\alpha}([g]) &= [\bar{\alpha}] \star [f] \star [\alpha] \star [\bar{\alpha}] \star [g] \star [\alpha] \\ &= [\bar{\alpha}] \star [f] \star [e_{x_0}] \star [g] \star [\alpha] \\ &= [\bar{\alpha}] \star [f] \star [g] \star [\alpha] \\ &= \hat{\alpha}([f] \star [g]) \end{aligned}$$

Hence $\hat{\alpha}$ is a homomorphism. To check that $\hat{\alpha}$ is an isomorphism, we see that $\hat{\alpha}(\hat{\alpha}([f])) = [f]$ for all $[f] \in \pi_1(X, x_0)$. □

Corollary 2.4.1. The fundamental group of X based at x_0 only depends on the path-connected component of $x_0 \in X$ upto isomorphism.

Definition 2.5. Let $h : X \rightarrow Y$ be a continuous map such that $h(x_0) = y_0$, $x_0 \in X, y_0 \in Y$. Then define $h_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$ by $h_*([f]) = [h \circ f]$. This is a **push-forward** map.

Proposition 2.6. h_* is a well-defined homomorphism.

Proof. Let $f \simeq_p f'$. We need to check that $h \circ f \simeq_p h \circ f'$. Let $H : I \times I \rightarrow X$ be the path homotopy between f and f' . Consider $h \circ H : I \times I \rightarrow Y$.

$$\begin{aligned} h \circ H(s, 0) &= h \circ f \\ h \circ H(s, 1) &= h \circ f' \\ h \circ H(0, t) &= h \circ f(0) \\ h \circ H(1, t) &= h \circ f(1) = h \circ f(0) \end{aligned}$$

Hence $h \circ H$ is the required path-homotopy.

Consider $[f], [g] \in \pi_1(X, x_0)$. We want to show that $h_*([f]) \star h_*([g]) = h_*([f] \star [g])$.

$$h_*([f]) \star h_*([g]) = [h \circ f] \star [h \circ g]$$

Note that it suffices to show that $h(f \star g) = (h \circ f) \star (h \circ g)$. Now,

$$\begin{aligned} h(f \star g)(s) &= \begin{cases} (h \circ f)(2s), & s \in [0, \frac{1}{2}] \\ (h \circ g)(2s - 1), & s \in [\frac{1}{2}, 1] \end{cases} \\ &= (h \circ f) \star (h \circ g) \end{aligned}$$

Hence proved. □

Theorem 2.7. If $h : (X, x_0) \rightarrow (Y, y_0)$ and $k : (Y, y_0) \rightarrow (Z, z_0)$ then

- (a) $(k \circ h)_* = k_* \circ h_*$
- (b) If $\text{id} : (X, x_0) \rightarrow (X, x_0)$ is the identity map then $(\text{id})_*$ is the identity $\pi_1(X, x_0) \rightarrow \pi_1(X, x_0)$.

Proof.

(a) We have,

$$\begin{aligned} (k \circ h)_*([f]) &= [(k \circ h) \circ (f)] \\ &= [k(h(f))] \\ &= [k_*(h_*([f]))] \\ &= (k_* \circ h_*)([f]) \end{aligned}$$

$$(b) (\text{id})_*[f] = [\text{id} \circ f] = [f]$$

□

Corollary 2.7.1. If $h : X \rightarrow Y$ is a homeomorphism then $\pi(X, x_0) \cong \pi(Y, h(x_0))$ via h_* .

Definition 2.8. Let $P : E \rightarrow B$ be a surjective map. A open set $U \subseteq B$ is said to be **evenly covered** by P if $P^{-1}(U) = \bigsqcup_{\alpha} V_{\alpha}$, where V_{α} are open sets of E for all α and $P|_{V_{\alpha}}$ is a homeomorphism of V_{α} onto U .

Definition 2.9. Let $P : E \rightarrow B$ be surjective if every point $b \in B$ has a neighbourhood $U \subset B$ such that U is evenly covered by P . Then we say that P is a **covering map** and E is a **covering space** of B .

Example 2.2.

(1) $\text{id} : X \rightarrow X$.

(2) $X \times \{1, 2, \dots, n\} \xrightarrow{P} X$ where $P(x, i) = x$.

Theorem 2.10. The map $P : \mathbb{R} \rightarrow S^1$ such that $x \mapsto (\cos(2\pi x), \sin(2\pi x))$ is a covering map.

Proof. Let $U_1 = S^1 \setminus \{(1, 0)\}$ and $U_2 = S^1 \setminus \{(-1, 0)\}$. Then it suffices to show that both of these sets is evenly covered. We have,

$$P^{-1}(U_1) = \sqcup_{n \in \mathbb{Z}} (n, n+1) =: V_n$$

And,

$$P^{-1}(U_2) = \sqcup_{n \in \mathbb{Z}} \left(n - \frac{1}{2}, n + \frac{1}{2} \right) =: W_n$$

Note that P is injective on any interval of length less than 1, because if $P(x) = P(y)$ then $(\cos(2\pi x), \sin(2\pi x)) = (\cos(2\pi y), \sin(2\pi y)) \iff x - y \in \mathbb{Z}$ but that is not possible if $x - y < 1$. $P|_{V_n}$ is surjective onto U_1 by IVT and is continuous because P is continuous. It has a continuous inverse since it is a continuous bijection from an open interval onto an open subset of S^1 .

Also, the intervals $(n, n+1)$ are pairwise disjoint for each n . A similar argument applies to W_n . Hence, both U_1 and U_2 are evenly covered and P is a covering map. $\square$

Definition 2.11. If a path-connected space X has $\pi_1(X, x_0) \cong 0$ then we say that X is **simply-connected**.

Lemma 2.12. If X is simply-connected, then any two paths in X with the same initial point and end point are path-homotopic.

Proof. $f \star \bar{g} \simeq_p e_{x_0}$ and $f \simeq_p f \star e_{x_1} \simeq_p f \star (\bar{g} \star g) \simeq_p (f \star \bar{g}) \star g \simeq_p e_{x_0} \star g \simeq_p g$. $\square$

Definition 2.13. Let $p : E \rightarrow B$ be a surjective map, and $f : X \rightarrow B$ be a map. Then a map $\tilde{f} : X \rightarrow E$ is called **lifting** of f to E if $p \circ \tilde{f} = f$.

$$\begin{array}{ccc} & & E \\ & \nearrow \tilde{f} & \downarrow p \\ X & \xrightarrow{f} & B \end{array}$$

Lemma 2.14 (Path Lifting Lemma). Let $p : E \rightarrow B$ be a covering map with $p(e_0) = b_0$. Any path $f : [0, 1] \rightarrow B$ beginning at b_0 can be lifted uniquely to a path $\tilde{f} : [0, 1] \rightarrow E$ beginning at e_0 .

Proof. Cover B by open sets U , each of which is evenly covered by p . We can find a subdivision (partition) defined by $s_0, \dots, s_n \in [0, 1]$ such that $f([s_i, s_{i+1}])$ is contained

entirely inside a evenly covered U for all i . Now we will define $\tilde{f}$ inductively. Suppose that $\tilde{f}(0) = e_0$ and it has been defined for $0 \leq s \leq s_i$ for some i . We will now define $\tilde{f}$ at $[s_i, s_{i+1}]$. Suppose that $f([s_i, s_{i+1}]) \subset U$ where U is evenly covered open set of B . This implies that $p^{-1}(U) = \sqcup_{\alpha} V_{\alpha}$. Now, $\tilde{f}(s_i)$ lies in one such V_{α} say V_0 .

Define $\tilde{f}$ on $[s_i, s_{i+1}]$ as $\tilde{f}(s) := (p|_{V_0})^{-1}(f(s))$ for all $s \in [s_i, s_{i+1}]$. By pasting lemma $\tilde{f}$ is continuous on $[0, s_{i+1}]$ and we argue inductively to define $\tilde{f}$ for $[0, 1]$.

By induction, we can again suppose that for another lift $\tilde{f}' : [0, 1] \rightarrow E$ of f with $\tilde{f}'(0) = e_0$, it agrees with $\tilde{f}$ of $0 \leq s \leq s_i$ for some i

$$\tilde{f}'(s) = \tilde{f}(s), \quad \forall s \in [0, s_i]$$

Now we would like to show that

$$\tilde{f}'(s) = \tilde{f}(s), \quad \forall s \in [s_i, s_{i+1}]$$

$$\tilde{f}'(s) = (p|_{V_0})^{-1} \circ f(s), \quad \forall s \in [s_i, s_{i+1}]$$

$\tilde{f}'(s_i)$ lies in V_0 since $\tilde{f}'(s_i) = \tilde{f}(s_i) \implies \tilde{f}'([s_i, s_{i+1}])$ lies in V_0 .

$p \circ \tilde{f}' = f$ on $[s_i, s_{i+1}]$. But $\tilde{f}'([s_i, s_{i+1}]) \subset V_0$ and $p|_{V_0}$ is a homeomorphism.

Then for all $s \in [s_i, s_{i+1}]$

$$\implies \tilde{f}'(s) = (p|_{V_0})^{-1} \circ f(s) = \tilde{f}(s)$$

□

Lemma 2.15 (Path Homotopy Lifting Lemma). *Let $p : E \rightarrow B$ be a covering map. Let $p(e_0) = b_0$. Let $f : I \times I \rightarrow B$ be a map, with $f(0,0) = b_0$. Then there exists a lifting of f to E such that $\tilde{f}(0,0) = e_0$. If f is a path homotopy, so is $\tilde{f}$.*

*Proof. **TODO:** $I \times I$ is a rectangle. Pick a partition of both sides, then the image of these small rectangles will be contained in U*

1. Define $\tilde{f}$ on $0 \times I$ and $I \times 0$ by lifting both of those paths to paths that begin at e_0 .
2. Take a subdivision of $I \times I$ say $I_i \times J_j$ such that each of those small rectangles are contained in an evenly covered open set of B .
3. By induction assume that $\tilde{f}$ has been defined on $I_i \times J_j$ such that $i < i_0, j = j_0$.
4. Now, to define $\tilde{f}$ on $I_{i_0} \times J_{j_0}$ we consider the image of the union of left and bottom edge of $I_{i_0} \times J_{j_0}$ under $\tilde{f}$. This lies inside $V_0 \subset p^{-1}(U)$. □

Remark 2.16. *The lifted path homomoty is also unique.*

Lemma 2.17. *Let $P : E \rightarrow B$ be a covering map with $P(e_0) = b_0$. Let f, g be two paths in B from b_0 to b_1 . Let $\tilde{f}, \tilde{g}$ be their lifts beginning at e_0 . Then if $f \simeq_p g \implies \tilde{f}$ and $\tilde{g}$ end at the same point and they are path homotopic.*

Proof. Assume $f \simeq_p g$ with the path homotopy $F : I \times I \rightarrow B$. Then Lemma 2.15 implies that F lifts to $\tilde{F} : I \times I \rightarrow E$ with $\tilde{F}(0, t) = e_0$.

Then we can say that $\tilde{F}(1, t) = e_1$ for some $e_1 \in P^{-1}(b_1)$, $\forall t \in I$. By uniqueness of path lifting, f, g have unique lifts to E , beginning at e_0 . We also know that $\tilde{F}(s, 0)$ and $\tilde{F}(s, 1)$ are lifts of f, g both beginning at e_0 .

Then $\tilde{F}(s, 0) = \tilde{f}$ and $\tilde{F}(s, 1) = \tilde{g}$ for all $s \in I$. Hence $\tilde{F}(1, 0) = \tilde{F}(1, 1) \implies \tilde{f}(1) = \tilde{g}(1)$, and $\tilde{f} \simeq \tilde{g}$. $\square$

Theorem 2.18. $\pi_1(S^1, (1, 0) = b_0) \cong (\mathbb{Z}, +)$

Proof.

$$\begin{array}{ccc} & & \mathbb{R} \\ & \nearrow \tilde{f} & \downarrow p \\ I & \xrightarrow{f} & S^1 \end{array}$$

Consider the covering map

$$P : \mathbb{R} \rightarrow S^1, P(x) = (\cos(2\pi x), \sin(2\pi x))$$

Let $[f] \in \pi_1(S^1, b_0)$. Let $\tilde{f}$ be the lift of f to $\mathbb{R}$ beginning at 0. Now consider $\tilde{f}(1)$. It will be an integer since $\tilde{f}(1) \in P^{-1}(b_0) = P^{-1}((1, 0))$. Define

$$\phi : \pi_1(S^1, b_0) \rightarrow \mathbb{Z}, \quad \phi([f]) = f(1)$$

By previous lemma, ϕ is well-defined.

Now we show that ϕ is surjective. Let $n \in \mathbb{Z}$. Define $\tilde{f}$ to be the path in $\mathbb{R}$ which starts at 0 and ends at n . Then by definition, $(P \circ \tilde{f})$ will start at $(1, 0)$ and end at $(1, 0)$ where $\tilde{f}$ is by definition the lift of $(P \circ \tilde{f})$ beginning at 0. So $\phi([P \circ \tilde{f}]) = n$.

To show that ϕ is injective suppose that $\phi([f]) = \phi([g]) = n$ for some $[f], [g] \in \pi_1(S^1, b_0)$ we need to show that $[f] = [g] \in \pi_1(S^1, b_0)$.

Now let $\tilde{f}$ and $\tilde{g}$ be two lifts of f and g respectively beginning at 0. Moreover, since $\mathbb{R}$ is simply connected and $\tilde{f}(0) = \tilde{g}(0) = 0, \tilde{f}(1) = \tilde{g}(1) = n, \tilde{f} \simeq_p \tilde{g}$, then $P \circ \tilde{F}$ will be the path-homotopy between f and g . This implies $[f] = [g]$.

Now we want to show that ϕ is a group homomorphism. Let $[f], [g] \in \pi_1(S^1, b_0)$ such that $\phi([f]) = n, \phi([g]) = m$ for $n, m \in \mathbb{Z}$. Now we want to show that $\phi([f] \star [g]) = \phi([f \star g]) = n + m$. We need to find a lift of $f \star g$ beginning at 0 and ending at $n + m$.

Let $\tilde{f}, \tilde{g}$ be lifts of f, g respectively, $\tilde{f}(0) = \tilde{g}(0) = (0)$.

Define a path on $\mathbb{R}$ by

$$h(s) = \begin{cases} \tilde{f}(2s), & s \in [0, \frac{1}{2}] \\ n + \tilde{g}(2s - 1), & s \in [\frac{1}{2}, 1] \end{cases}$$

Then h is a path beginning at 0 and ending at $n + m$. Now consider,

$$P(h(s)) = \begin{cases} P \circ \tilde{f}(2s), & s \in [0, \frac{1}{2}] \\ P \circ (n + \tilde{g}(2s - 1)) = P \circ \tilde{g}(2s - 1), & s \in [\frac{1}{2}, 1] \end{cases}$$

This implies that $h(s)$ is a lift of $f \star g(s) \implies \phi([f \star g]) = h(1) = n + m$ $\square$

Corollary 2.18.1. *Fundamental Theorem of Algebra.*

Proof. Let $p(z) = z^n + a_1 z^{n-1} + \dots + a_n$, where $a_i \in \mathbb{C}$.

For the sake of contradiction assume that p has no roots in $\mathbb{C}$. Then for each $r \in \mathbb{R}^+$ define a function

$$f_r(s) = \frac{\frac{p(re^{2\pi i s})}{p(r)}}{\left| \frac{p(re^{2\pi i s})}{p(r)} \right|} \in \mathbb{C}$$

Note that $f_r(s)$ is a loop in S^1 based at $(1, 0)$ for all r . ($f_r(0) = (1, 0) = f_r(1)$ and $|f_r(s)| = 1, \forall s$). Note that as r varies f_r gives a path homotopy in S^1 between loops that are based at $(1, 0)$. Now, note that $f_0(s)$ is the trivial loop based at $(1, 0)$. This implies,

$$[f_r(s)] = [f_0(s)] = 0 \in \pi_1(S^1, (1, 0))$$

Now fix a large value of r such that $r > |a_1| + \dots + |a_n|$ and $r > 1$. Now, we have for $|z| = r$

$$|a_1 z^{n-1} + \dots + a_n| \leq |a_1| r^{n-1} + \dots + |a_n|.$$

Thus if $r > |a_1| + \dots + |a_n|$, then

$$|z^n| = r^n > r^{n-1}(|a_1| + \dots + |a_n|) \geq |a_1 z^{n-1} + \dots + a_n|.$$

Hence,

$$|z^n| > |a_1 z^{n-1} + \dots + a_n|$$

This implies that the polynomial $p_t(z) = z^n + t(a_1 z^{n-1} + \dots + a_n)$, $\forall t \in [0, 1]$ has no roots in circle $|z| = r$. If we now replace p by p_t in the definition of $f_r(s)$ we get

$$f_r(s, t) = \frac{\frac{p_t(re^{2\pi i s})}{p_t(r)}}{\left| \frac{p_t(re^{2\pi i s})}{p_t(r)} \right|}$$

At $t = 1$,

$$f_r(s, 1) = f_r(s) \in \pi_1(S^1, (1, 0))$$

At $t = 0$,

$$f_r(s, 0) = e^{2\pi i n s}, \quad \forall s \in [0, 1]$$

So $[f_r(s, 0)] = n \in \mathbb{Z} \cong \pi_1(S^1, (1, 0))$. But $f_r(s, t)$ as we vary $t \in [0, 1]$ gives a path homotopy between $f_r(s, 0)$ and $f_r(s, 1) \implies [f_r(s, 0)] = [f_r(s)] \in \pi_1(S^1, (1, 0)) \implies n = 0$ contradicting that p is not constant. $\square$

Remark 2.19. In general, we have $P : (E, e_0) \rightarrow (B, b_0)$ be a covering map. If E is path-connected then there exists a surjection $\phi : \pi_1(B, b_0) \rightarrow P^{-1}(b_0)$. If E is simply-connected, then ϕ is a bijection.

Proof.

(a) For $[\gamma] \in \pi_1(B, b_0)$, let $\gamma' : I \rightarrow E$ be the unique lift of γ starting at e_0 . Define

$$\phi : \pi_1(B, b_0) \rightarrow p^{-1}(b_0), \quad \phi([\gamma]) = \gamma'(1).$$

Note $\gamma'(1) \in p^{-1}(b_0)$ since $p(\gamma'(1)) = \gamma(1) = b_0$, so ϕ is well-defined.

To see surjectivity, let $e_\alpha \in p^{-1}(b_0)$. Since E is path-connected, there exists a path $\sigma_\alpha : I \rightarrow E$ such that $\sigma_\alpha(0) = e_0$ and $\sigma_\alpha(1) = e_\alpha$.

Then $\gamma = p \circ \sigma_\alpha$ is a loop in B based at b_0 , since

$$\gamma(0) = (p \circ \sigma_\alpha)(0) = p(e_0) = b_0 \quad \text{and} \quad \gamma(1) = (p \circ \sigma_\alpha)(1) = p(e_\alpha) = b_0$$

By uniqueness of lifts, σ_α is the unique lift of γ starting at e_0 , so $\phi([\gamma]) = \sigma_\alpha(1) = e_\alpha$. We can similarly define a unique lift for each e_α and thus ϕ is surjective.

(b) By part (a) we already know that ϕ is surjective. To see that ϕ is injective, let $[f], [g] \in \pi_1(B, b_0)$ such that $\phi([f]) = \phi([g]) = e_\alpha$ for some $e_\alpha \in p^{-1}(b_0)$, and let f', g' be the lifts of f, g starting at e_0 .

Then $f'(0) = g'(0) = e_0$ and $f'(1) = g'(1) = e_\alpha$. Since E is simply connected, $\pi_1(E, e_0) \cong \{e\}$, i.e. any two paths with the same initial and final points in E are path-homotopic. Thus, $[f'] = [g']$, let F' be the homotopy between f' and g' . Then we have

$$F'(s, 0) = f'(s) \quad \text{and} \quad F'(s, 1) = g'(s).$$

Applying p to both sides,

$$(p \circ F')(s, 0) = p(f'(s)) = f(s) \implies p(f') = f$$

$$(p \circ F')(s, 1) = p(g'(s)) = g(s) \implies p(g') = g$$

Therefore $p \circ F' : I \times I \rightarrow B$ is a homotopy between f and g . Hence, $[f] = [g]$, ϕ is injective, and thus bijective. □

Theorem 2.20.

$$\pi_1(\mathbb{R}^2 \setminus \{(0, 0)\}, b_0) \cong \mathbb{Z} \cong \pi_1(S^1), \quad b_0 \in S^1.$$

Proof. Let $x_0 \in S^1$. Then the inclusion map

$$J : (S^1, b_0) \rightarrow (\mathbb{R}^2 \setminus \{(0, 0)\}, b_0)$$

induces an isomorphism

$$J_* : \pi_1(S^1, b_0) \rightarrow \pi_1(\mathbb{R}^2 \setminus \{(0, 0)\}, b_0).$$

Consider

$$r : \mathbb{R}^2 \rightarrow S^1, \quad r(x) = \frac{x}{\|x\|}.$$

Observe that $r \circ J = \text{id}$, thus

$$r_* \circ J_* = \text{id}.$$

Thus J_* is injective.

Choose $[f] \in \pi_1(\mathbb{R}^2 \setminus \{(0,0)\}, b_0)$. Recall that,

$$J_* \circ r_*([f])$$

is explicitly $J \circ r \circ f$. Let

$$g(s) = \frac{f(s)}{\|f(s)\|}.$$

Next show $g \simeq_p f$. Define

$$F : I^2 \rightarrow \mathbb{R}^2 \setminus \{(0,0)\}$$

such that

$$F(s, t) = t \cdot \frac{f(s)}{\|f(s)\|} + (1-t)f(s).$$

F is the straight line homotopy and does not cross 0.

Since $f(s) \neq 0$ by definition, suppose

$$t \cdot \frac{1}{\|f(s)\|} + (1-t) = 0$$

implies $\|f(s)\|$ negative or $t \notin [0, 1]$.

Thus F is well-defined, since $f \simeq_p g$,

$$J_* \circ r_*([f]) = [f].$$

Thus surjection. □

3. RETRACTIONS

Definition 3.1. If A is a subspace of X s.t. $\exists$ a map

$$r : X \rightarrow A$$

with $r(a) = a$ for all $a \in A$, then we say A is a retract of X and r is a retraction map from X to A .

Definition 3.2. Let A be a subspace of X . A is a deformation retract of X if there exists a homotopy

$$H : X \times I \rightarrow X$$

such that

$$H(x, 0) = x \quad \forall x \in X,$$

$$H(x, 1) \in A,$$

$$H(a, t) = a.$$

Theorem 3.3. Let A be a deformation retract of X . Then $\iota : (A, x_0) \hookrightarrow (X, x_0)$ induces an isomorphism in $\pi_1(A, x_0)$ to $\pi_1(X, x_0)$. If A is a retract of X , then ι is injective.

Proof.

- (a) Let r be the map that retracts X to A , and $r(x_0) = a_0$ in A . Note that the composition $r \circ \iota$ is the identity on A . Now consider the homomorphism $(r \circ \iota)_*$. We have,

$$(r \circ \iota)_* = r_* \circ \iota_* = (\text{id}_A)_* = \text{id}_{\pi_1(A, a_0)}$$

Let $\iota_*([f]) = \iota_*([g])$. Then,

$$\begin{aligned} r_*(\iota_*([f])) &= r_*(\iota_*[g]) \\ \implies (\text{id}_A)_*([f]) &= (\text{id}_A)_*([g]) \\ \implies [f] &= [g] \end{aligned}$$

Hence ι_* is injective.

- (b) Let H be the deformation retraction of X to A , and $H(x_0, 1) = a_0$ in A . From part (a) we know that ι_* is injective. So we only need to show that ι_* is surjective. We define the continuous map $r = H(x, 1)$. Consider the map $\iota \circ r$ which maps $X \rightarrow A \rightarrow X$. This map is homotopic to the identity map by H (by definition),

$$\iota \circ r \simeq \text{id}_X$$

Then $(\iota \circ r)_* = \iota_* \circ r_* \simeq (\text{id}_X)_* = \text{id}_{\pi_1(X, x_0)}$ (Lemma 58.1). Since $\text{im}(r_*) = \pi_1(A, a_0)$, ι_* is surjective. Thus, we have an isomorphism $\pi_1(A, a_0) \cong \pi_1(X, x_0)$.

□

4. APPLICATIONS OF π_1 -COMPUTATIONS

Theorem 4.1 (Brouwer-Fixed Point Theorem). *Every map $h : D^2 \rightarrow D^2$ has a fixed point, i.e. $h(x) = x$ for at least one $x \in D^2$.*

Proof. Suppose towards contradiction $h(x) \neq x$ for all $x \in D^2$. Define a function $\gamma : D^2 \rightarrow S^1$ as follows: consider the straight line starting at $h(x)$ and going to x . Define $r(x)$ as the intersection of that line with $\partial D^2 = S^1$. Now r is a continuous function, and $r|_{S^1} = Id_{S^1}$. That is, r is a retraction of D^2 to S^1 .

Since for the inclusion $\iota : S^1 \hookrightarrow D^2$ we should have $\pi_1(S^1, (1,0)) \xrightarrow{\iota_*} \pi_1(D^2, (1,0))$ is an injection which gives a contradiction. $\square$

Theorem 4.2 (Borsuk-Ulam Theorem). *Every map $f : S^2 \rightarrow \mathbb{R}^2$ has a pair of anti-podal points (x is an antipode of $-x$, $x \in S^2$) in S^2 such that $f(x) = f(-x)$.*

Proof. Suppose that this is false. Then $f(x) \neq f(-x)$ for all $x \in S^2$. We define $g : S^2 \rightarrow S^1$ by

$$g(x) = \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|}$$

Define a loop η by the equator that lies in the $X - Y$ plane of S^2 .

$$\eta(s) = (\cos(2\pi s), \sin(2\pi s), 0), s \in [0, 1]$$

and let $h : I \rightarrow S^1$ defined by $h := g \circ \eta$. Note that since $g(x) = -g(-x)$ we have $h(s + \frac{1}{2}) = -h(s)$ for all $s \in [0, \frac{1}{2}]$.

$$\begin{aligned} g(\cos(2\pi(s + 1/2)), \sin(2\pi(s + 1/2)), 0) &= g(-\cos(2\pi s), -\sin(2\pi s), 0) \\ &= -g(\cos(2\pi s), \sin(2\pi s), 0) \end{aligned}$$

By path-lifting lemma, we can lift h to a path in $\mathbb{R}$, say $\tilde{h}$ starting at 0 such that

$$\tilde{h}\left(s + \frac{1}{2}\right) = \tilde{h}(s) + \frac{q(s)}{2}$$

for some odd integer values $q(s)$. But now, $q(s) = 2(\tilde{h}(s + \frac{1}{2}) - \tilde{h}(s))$ and hence it is also a continuous function. This implies $q(s)$ is a constant map (since the domain of q is connected), i.e. $\tilde{h}(s + \frac{1}{2}) = \tilde{h}(s) + \frac{q}{2}$.

$$\tilde{h}(1) = \tilde{h}\left(\frac{1}{2}\right) + \frac{q}{2} = \tilde{h}(0) + q$$

This implies $\tilde{h}(1) = q$ and hence $[h] \in \pi_1(S^1, (1,0))$ is not homotopic to the identity. But since η is nullhomotopic in S^2 , $g \circ \eta$ is homotopic to a constant map $[h] = 0 \in \pi_1(S^1, (1,0))$. A contradiction! $\square$

5. HOMOTOPY EQUIVALENCES

Theorem 5.1. $\pi_1(X \times Y, x_0 \times y_0) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$ for any pair of topological spaces X and Y .

Proof. Let $P_1 : X \times Y \rightarrow X$ and $P_2 : X \times Y \rightarrow Y$ be the respective projections.

$$\phi : \pi_1(X \times Y, x_0 \times y_0) \rightarrow \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

where

$$\phi([f]) = (P_1)_*[f] \times (P_2)_*[f]$$

ϕ is surjective: Let $[g] \in \pi_1(X, x_0)$ and $[h] \in \pi_1(Y, y_0)$. Then define $[g] \times [h] : I \rightarrow X \times Y$ by $s \mapsto g(s) \times h(s)$. Then $\phi([g] \times [h]) = [g] \times [h]$.

ϕ is injective: Suppose that $[f] \in \pi_1(X \times Y, x_0 \times y_0)$ such that $\phi([f]) = (P_1)_*([f]) \times (P_2)_*([f]) = 0 = ([e_{x_0}] \times [e_{y_0}])$. Then $P_1 \circ f \simeq_p e_{x_0}$ and $P_2 \circ f \simeq_p e_{y_0}$ by homotopies F_t and G_t .

Then, $F_t \times G_t : I \rightarrow X \times Y$ gives $f \simeq_p e_{x_0 \times y_0} \implies [f] = 0$. □

Definition 5.2. We say $f : X \rightarrow Y$ is a homotopy equivalence if there exists $g : Y \rightarrow X$ such that $f \circ g \simeq id_Y$ and $g \circ f \simeq id_X$. Then we say that X is homotopy equivalent to Y and write it as $X \simeq_h Y$.

Remark 5.3. Homotopy equivalence is an equivalence relation on topological spaces.

Proposition 5.4. Suppose that X deformation retracts to A . Then $X \simeq_h A$.

Proof. Let $H : X \times I \rightarrow A$ be the deformation retraction, and $\iota : A \hookrightarrow X$ be the inclusion map. Then let $r = H(x, 1)$ for all $x \in X$. Now note that,

$$r \circ \iota \simeq id_A$$

And, by definition of the deformation retraction H gives us,

$$\iota \circ r \simeq id_X$$

Thus, $X \simeq_h A$. □

Remark 5.5. Note not all homotopy equivalence come from deformation retractions.

Remark 5.6. π_1 is not only an invariant of homeomorphism class of a topological space but also an invariant of the homotopy equivalence class of the space.

Lemma 5.7. If $\phi_t : X \rightarrow Y$ be a homotopy and h is the path given by $\phi_t(x_0)$ for a choice of $x_0 \in X$, then we have $(\phi_0)_* = \widehat{h}((\phi_1)_*)$ on $\pi_1(X, x_0)$. That is,

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{(\phi_1)_*} & \pi_1(Y, \phi_1(x_0)) \\ & \searrow (\phi_0)_* & \downarrow \widehat{h} \\ & & \pi_1(Y, \phi_0(x_0)) \end{array}$$

Proof. Let $[f] \in \pi_1(X, x_0)$. Let h_t be a reparameterization of the path h such that $h_t(s) := h(st)$. In particular, h_t is a path from $\phi_0(x_0)$ to $\phi_t(x_0)$. Now the composition, $h_t \star \phi_t \circ f \star \bar{h}_t$ gives a path homotopy between

$$h_0 \star \phi_0 \circ f \star \bar{h}_0 = e_{x_0} \star \phi_0 \circ f \star \bar{e}_{x_0} \simeq \phi_0 \circ f$$

and

$$h_1 \star \phi_1 \circ f \star \bar{h}_1 = h \star \phi_0 \circ f \star \bar{h} \implies (\phi_0)_* = \widehat{h}((\phi_1)_*)$$

□

Theorem 5.8. *Let $\phi : X \rightarrow Y$ be a homotopy equivalence. Then $\phi_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, \phi(x_0))$ is an isomorphism.*

Proof. Since ϕ is a homotopic equivalence, there exists a $\psi : Y \rightarrow X$ such that $\phi \circ \psi \simeq \text{id}_Y$ and $\psi \circ \phi \simeq \text{id}_X$. Consider,

$$\pi_1(X, x_0) \xrightarrow{\phi_*} \pi_1(Y, \phi(x_0)) \xrightarrow{\psi_*} \pi_1(X, \psi(\phi(x_0))) \xrightarrow{\phi_*} \pi_1(Y, \phi(\psi(\phi(x_0))))$$

Note that $\psi_* \circ \phi_* = (\psi \circ \phi)_* = \widehat{h} \circ (\text{id}_X)_*$ for some path h in X (by the previous lemma).

This implies $\psi_* \circ \phi_* = \widehat{h} \implies \psi_*$ is surjective since $\widehat{h}$ is an isomorphism.

ψ_* is injective because

$$\phi_* \circ \psi_* = \widehat{h'} \circ (\text{id}_Y)_*$$

for some path h' in Y (by the previous lemma).

Hence, $\phi_* = (\psi_*)^{-1} \circ \widehat{h}$ which is an isomorphism.

□

6. $\pi_1(S_n)$

Lemma 6.1. *If a space X is the union of a collection of path-connected open sets A_α each containing the base point $x_0 \in X$. Let $A_\alpha \cap A_\beta$ be path connected for any α, β , then every loop in X based at x_0 is path homotopic to a composition of loops each contained in a single A_α for some α and based at x_0 .*

Proof. Let $[f] \in \pi_1(X, x_0)$. Suppose that $0 = S_0 < S_1 < \dots < S_m = 1$ is a subdivision of $[0, 1]$ such that each interval $[S_{i-1}, S_i]$ is mapped by f to a single A_α , let f_i be the path obtained by the restriction $f|_{[S_{i-1}, S_i]}$. Let us denote that A_α by A_i . Note $f(0) = x_0$. Define g_i to be the path from x_0 to $f(S_i) \in A_i \cap A_{i+1}$.

Consider,

$$(f_1 \star \overline{g_1}) \star (g_1 \star f_2 \star \overline{g_2}) \star (g_2 \star f_3 \star \overline{g_3}) \star \dots \star (g_{m-1} \star f_m) \simeq_p f$$

while each of them are loops based at x_0 in A_i s as i varies. □

Theorem 6.2. $\pi_1(S^n, x_0) \cong \{0\}$ if $n \geq 2$.

Proof. On S^2 consider two open sets $A_1 = S^2 \setminus \{\text{north pole}\}$ and $A_2 = S^2 \setminus \{\text{south pole}\}$. Let $[f] \in \pi_1(S^2, x_0)$. Then by the lemma $f \simeq_p g_1 \star g_2$ where $[g_1] \in \pi_1(A_1, x_0)$ and $[g_2] \in \pi_1(A_2, x_0)$ for some g_1 and g_2 loops based at x_0 . But since $\pi_1(A_1) = \pi_1(A_2) = 0$,

$$[f] = 0 \in \pi_1(S^2, x_0) \implies \pi_1(S^2, x_0) = 0$$

(Same proof for any $n \geq 2$). □

Remark 6.3 (Stereographic Projection). $S^n \setminus \{\text{some pt}\}$ is homeomorphic to $\mathbb{R}^n$.

7. COVERING SPACES (AGAIN)

Remark 7.1. For every group G there exists a space X_G such that $\pi_1(X_G) \cong G$.

Definition 7.2. $\mathbb{RP}^n$ can be defined the following ways:

- (1) The quotient of $\mathbb{R}^{n+1} \setminus \{0\}$ under the equivalence relation $x \sim \lambda x$ for $0 \neq \lambda \in \mathbb{R}$.
- (2) The quotient of S^n under the equivalence relation $x \sim -x$.

Theorem 7.3. Define $p : S^n \rightarrow \mathbb{RP}^n, x \mapsto [x]$. Then p is a covering map.

Proof. First show that p is an open map (i.e. U open $\implies p(U)$ open). Let $V \subset S^n$ be an open set (S^n has subspace topology from $\mathbb{R}^{n+1}$). Consider ‘antipodal map’,

$$a : S^n \rightarrow S^n, \quad x \mapsto -x$$

Then a is a homeomorphism. Thus its inverse is continuous, and $a(V)$ is open in S^n . Now since $p^{-1}(p(V)) = V \cup a(V)$, which is open in S^n , p is an open map (using definition of quotient topology).

Now we will show p is a covering map. Let $y \in \mathbb{RP}^n$. Pick $x \in p^{-1}(y)$. Choose a small neighborhood U of x in S^n such that U contains no antipodal points. Then $p|_U : U \rightarrow p(U)$ is a homeomorphism (since p is open and continuous and a bijection).

Similarly, $p|_{a(U)} : a(U) \rightarrow p(a(U))$ is also an homeomorphism. Note $p(a(U)) = p(U)$ which is an open set around $y \in \mathbb{RP}^n$. So each point in $\mathbb{RP}^n$ is evenly covered by an open set in $\mathbb{RP}^n$. Hence, p is a covering map (specifically a 2-fold covering).

Then $\pi_1(\mathbb{RP}^n, x_0) \hookrightarrow p^{-1}(x_0) \implies |\pi_1(\mathbb{RP}^n, x_0)| = 2 \implies |\pi_1(\mathbb{RP}^n, x_0)| = \mathbb{Z}/2\mathbb{Z}$. □

Corollary 7.3.1. $\mathbb{RP}^n$ is not homotopy equivalent to S^n for $n \geq 2$.

Proof. Fundamental group is a homotopy-equivalence invariant, and S^n for $n \geq 2$ is simply connected. □

Remark 7.4. Homeomorphism $X \cong Y \implies X \simeq_h Y$ (homotopy equivalence).

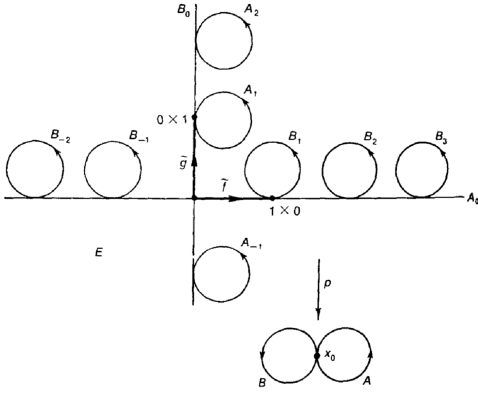
Theorem 7.5. $\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n \neq 2$.

Proof. Suppose for contradiction there exists a homeomorphism f from $\mathbb{R}^2 \rightarrow \mathbb{R}^n$. Then $f|_{\mathbb{R}^2 \setminus \{pt\}} : \mathbb{R}^2 \setminus \{x\} \rightarrow \mathbb{R}^n \setminus \{f(x)\}$ is also a homeomorphism. So $\mathbb{R}^2 \setminus \{x\} \simeq_h \mathbb{R}^n \setminus \{f(x)\}$ must be homotopy equivalent, But π_1 is a homotopy equivalence so we should have $\pi_1(\mathbb{R}^2 \setminus \{x\}) \cong \pi_1(S^1) \cong \pi_1(\mathbb{R}^n \setminus \{f(x)\}) \implies \mathbb{Z} \cong 0$. A contradiction! □

Theorem 7.6. Fundamental group of the ‘figure eight space’ is not abelian.

Proof. Let X be the figure eight space. Then $X = S^1 \sqcup S^1 / (1, 0) \sim (1, 0)$ with the quotient space topology. Let x_0 be the base point. Define a covering space $E \subseteq \mathbb{R}^2$, where

$$E = \{x\text{-axis}\} \cup \{y\text{-axis}\} \cup \{\text{a circle tangent to both axes at each nonzero integer point}\}$$



We claim that there exists a projection map $p : E \rightarrow X$. Consider two paths $\tilde{f}, \tilde{g} : I \rightarrow E$ defined by $\tilde{f}(s) = (s, 0)$, $\tilde{g}(s) = (0, s)$. Define $f := p \circ \tilde{f}$, $g := p \circ \tilde{g}$. That is f wraps once around A , g wraps once around B .

Claim: $[f \star g] \neq [g \star f] \in \pi_1(\infty, x_0)$.

Proof: Claim $\iff f \star g \not\simeq_p g \star f$. For the sake of contradiction suppose $f \star g \simeq_p g \star f$. Now consider the lift of $f \star g$ beginning at $(0, 0)$ to E given by $\tilde{f} \star B_1$, and the lift of $g \star f$ beginning at $(0, 0)$ to E given by $\tilde{g} \star A_1$. If they are path-homotopic, they should have the same end points, but $(\tilde{f} \star B_1)(1) = (1, 0) \neq (0, 1) = (\tilde{g} \star A_1)(1)$. $\square$

Corollary 7.6.1. $\pi^2 \# \pi^2 \not\cong S^2, \mathbb{RP}^2$, where π^2 is the hollow torus, and $\#$ is the connected sum.
TODO: ??

Proposition 7.7. Let $h : S^1 \rightarrow Y$ where Y is some topological space, then h is homotopic to a constant map iff h can be extended to a continuous map $D^2 \rightarrow Y$.

Proof. ($\implies$)

Let $H : S^1 \times I \rightarrow Y$ be a homotopy from h to a constant map $S^1 \rightarrow x_0$. Define $\pi : S^1 \times I \rightarrow D^2$ by $\pi(x, t) = (1 - t)x$. Then $\pi|_{S^1 \times [0, 1)}$ is a bijection to $D^2 \setminus \{0\}$ and it maps $S^1 \times \{1\}$ to $0 \in D^2$.

Since $S \times I$ is compact and D^2 is Hausdorff, π is a closed map. Now H induces a map $g : D^2 \rightarrow Y$ such that $g \circ \pi = H$, given by

$$g(x) = \begin{cases} H(\pi^{-1}(x)) & x \neq 0 \\ y_0 & x = 0 \end{cases}$$

The map g is continuous since π is a quotient map. Thus the map g is the desired extension of h for if $x \in S^1$ then,

$$g(x) = g(\pi(x, 0)) = H(x, 0) = h(x)$$

($\impliedby$)

Let $g : D^2 \rightarrow Y$ be a continuous extension of h , and define $F : S^1 \times I \rightarrow Y$ by

$$F(x, t) = g((1 - t)x)$$

Then F is a homotopy between h and a constant map. $\square$

TODO: 05/04 notes

Remark 7.8. By the Galois correspondence, covers of X are in bijection with subgroups of $\pi_1(X)$.

Proposition 7.9. Let $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a covering map. Then the map p_* is injective. Moreover, $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ consists of homotopy classes of loops in X that lift to loops in $\tilde{X}$ based at $\tilde{x}_0$.

Proof. Let $[f] \in \pi_1(\tilde{X}, \tilde{x}_0)$ and suppose $p_*([f]) = 0 \in \pi_1(X, x_0)$. Then there exists a path homotopy H between $p \circ f$ and the constant loop c_{x_0} . By the homotopy lifting lemma, we can lift this to a homotopy between f and the trivial loop $c_{\tilde{x}_0}$ in $\tilde{X}$. Thus, $[f] \simeq_p 0$, and p_* is injective.

Moreover, loops at x_0 lifting to loops at $\tilde{x}_0$ certainly represent elements of $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$, and conversely, a loop representing an element of $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ is homotopic to a loop having such a lift, so by homotopy lifting, the loop itself must have such a lift. $\square$

Definition 7.10. Given $p : \tilde{X} \rightarrow X$ a covering, $|p^{-1}(x)|$ for $x \in X$ is the number of **sheets**.

Proposition 7.11. Given X and $\tilde{X}$ path connected, then the number of sheets of $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ of the covering equals the index of $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ in $\pi_1(X, x_0)$.

Proof. Let g be a loop based at x_0 in X . Let $\tilde{g}$ be its lift in $\tilde{X}$, starting at $\tilde{x}_0$.

Now let $[h] \in H := p_*(\pi_1(\tilde{X}, \tilde{x}_0))$, and let $\tilde{h} \circ \tilde{g}$ be a lift based at $\tilde{x}_0$, starting at $\tilde{x}_0$. We look at the lift of $h \circ g$, $\tilde{h} \circ \tilde{g}$ which ends at the same point as $\tilde{g}$ since $\tilde{h}$ is a loop at $\tilde{x}_0$.

Thus we can define a function ϕ from the cosets $H[g]$ to $p^{-1}(x_0)$. ϕ is surjective since $\tilde{X}$ is path-connected and hence $\tilde{x}_0$ can be joined to any point in $p^{-1}(x_0)$ by a path $\tilde{g}$ that projects to a loop g at x_0 .

To see that ϕ is injective. Let $\phi(H[g_1]) = \phi(H[g_2])$. Then, the lifts corresponding to g_1 and g_2 end at the same point, and thus $[g_1 \circ \bar{g}_2] = [g_1][g_2]^{-1} \in H \implies H[g_1] = H[g_2]$.

Thus there exists a bijection between the set of cosets of $\pi_1(X, x_0)$ and the fibre $p^{-1}(x_0)$ and $|p^{-1}(x_0)| = \text{number of cosets} = [\pi_1(X, x_0) : H]$. $\square$

Proposition 7.12. Given a covering space $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ and a map $f : (Y, y_0) \rightarrow (X, x_0)$ with $p(\tilde{x}_0) = x_0$ and $f(y_0) = y_0$ with Y path-connected and locally path-connected, then there exists a lift $\tilde{f} : (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ of f iff $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.

Proof. ($\implies$)

Assume there exists a lift $\tilde{f}$ then we know that $f_* = p_* \circ \tilde{f}_*$, and thus

$$\text{im}(f_*) = \text{im}(p_* \circ \tilde{f}_*) \subseteq \text{im}(p_*) \implies f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0))$$

($\impliedby$)

Assume $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0))$. Let $y \in Y$ and γ a path in Y from y_0 to y . Let $\tilde{f}\gamma$ be the unique lift of the path $f \circ \gamma$ in X starting at $\tilde{x}_0$. Now define $\tilde{f} : (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$

$$\tilde{f}(y) = \tilde{f}\gamma(1)$$

This is well-defined, let δ be some arbitrary path from y_0 to y . Then $f_\gamma \cdot \overline{f_\delta}$ is a loop based at x_0 .

$$[f_\gamma] \cdot [f_\delta]^{-1} \in f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0)), \quad \text{by hypothesis.}$$

Thus there exists a path homotopy from f_γ to f_δ to a loop in X which lifts to a loop in $\tilde{X}$, based at $\tilde{x}_0$. **TODO: details idrc**

Now we prove continuity of $\tilde{f}$. Let $y \in Y$. Choose $U \subseteq X$ that is an openly covered neighbourhood of $f(y)$ having a lift $\tilde{U}$ such that $p : \tilde{U} \rightarrow U$ is a homeomorphism. Also pick a path-connected open neighbourhood V of y s.t. $f(V) \subseteq U$.

Choose $y' \in V$. Let δ be a path from y_0 to y and η a path from y to y' in V . Thus $f_\delta \cdot f_\eta$ lifts to $\tilde{f}_\delta \cdot \tilde{f}_\eta$, where $\tilde{f}_\eta = p^{-1}f_\eta$ and $p^{-1} : U \rightarrow \tilde{U}$. Thus, $\tilde{f}(V) \subset \tilde{U}$ and $\tilde{f}|_V = p^{-1}f$. Hence $\tilde{f}$ is continuous at y . $\square$

8. CLASSIFICATION OF COVERING SPACE

Definition 8.1. We say that a covering space $\tilde{X}$ of X is a universal cover if $\pi_1(\tilde{X}) = 0$.

Example 8.1. (1) $\mathbb{R}$ is a universal cover of S^1 .

(2) Every simply connected space is a universal cover of itself.

(3) $\mathbb{C} \rightarrow \mathbb{C}$ where $z \mapsto z^2$.

(4) S^n is a universal cover of $\mathbb{RP}^n$ for $n \geq 2$

Remark 8.2. An object with property x is universal if any other object with property x factors through the first object.

Theorem 8.3. Suppose that $\tilde{X}$ is a universal cover of X , then if X is path-connected and locally-connected, any other path-connected cover X_1 of X , $\tilde{X}$ covers X_1 . Moreover the following diagram commutes.

$$\begin{array}{ccccc} \tilde{X} & \xrightarrow{\quad p_1 \quad} & X_1 & \xrightarrow{\quad p_2 \quad} & X \\ & \searrow & \uparrow & \nearrow & \\ & & p & & \end{array}$$

Proof. The proof follows from the following two lemmas. □

Lemma 8.4. Suppose that $X, X_1, \tilde{X}$ are all path-connected such that $X_1 \xrightarrow{p_2} X$, $\tilde{X} \xrightarrow{p} X$ are covers with $\pi_1(\tilde{X}) = 0$ then $\exists p_1 : \tilde{X} \rightarrow X_1$ such that $p_1 \circ p_2 = p$.

Proof. Follows from the general lifting proposition. □

Lemma 8.5. Let $X, X_1, \tilde{X}, p, p_1, p_2$ be defined as before, then p_1 is a covering map.

Proof. Let $x_1 \in X_1$ be a point, U be a neighborhood of $p_2(x_1) \in X$ that is evenly covered by both p_2 and p . Now let $p^{-1}(U) = \sqcup_{\alpha} U_{\alpha}$ and $p_2^{-1}(U) = \sqcup_{\alpha'} V_{\alpha'}$. Moreover assume that V is in the slice of $p_2^{-1}(U)$ where x_1 lies.

Claim: V is evenly covered by p_1 .

Proof: Consider the U_{α} 's in $\tilde{X}$ that are mapped to V . Lets say that collection is $\{U_{\beta}\}$. Then $p_1^{-1}(V) = \sqcup_{\beta} U_{\beta}$ and $p_1|_{U_{\beta}}$ is a homeomorphism since it is a composition of $p|_{U_{\beta}}$ and $(p_2|_V)^{-1}$ which are both homeomorphisms □

Theorem 8.6. Let $\tilde{X}$ be a universal cover of X , then $\tilde{X}$ is unique upto homeomorphism.

Proof. Hint: Lifts are unique if we fix a base point of the lift.

Let $\tilde{X}$ and $\bar{X}$ be two universal covers of X . Then,

$$\begin{array}{ccc} & & (\bar{X}, \bar{x}_0) \\ & \nearrow f & \downarrow p_2 \\ (\tilde{X}, \tilde{x}_0) & \xrightarrow{p_1} & (X, x_0) \end{array} \quad .$$

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$$\begin{array}{ccc}
& & (\tilde{X}, \tilde{x}_0) \\
& \nearrow g & \downarrow p_1 \\
(\bar{X}, \bar{x}_0) & \xrightarrow{p_2} & (X, x_0)
\end{array}$$

$$p_1 = p_2 \circ f, \quad p_2 = p_1 \circ g$$

That is,

$$p_1 = p_1 \circ g \circ f$$

Moreover, $g \circ f(\tilde{x}_0) = \tilde{x}_0$. Hence, $g \circ f$ is a lift of p_1 to $\tilde{X}$ by p_1 . Identity is also such a lift, then $g \circ f = \text{id}$ (by uniqueness of lift homotopy). $\square$

TODO: Read this proof ?? For any sub space of $\pi_1(X) \leftrightarrow$ cover X_1 of X with $p_*(\pi_1(X_1)) \cong$ that subspace.

Remark 8.7. Any collection of homomorphisms $\phi_\alpha : G_\alpha \rightarrow H$ extends uniquely to $\phi : \star_\alpha G_\alpha \rightarrow H$ such that $\phi(g_1 \cdots g_n) \mapsto \phi_{\alpha_1}(g_1)\phi_{\alpha_2}(g_2) \cdots \phi_{\alpha_n}(g_n)$ where G_α is a free group, H is a group, and $\star_\alpha G_\alpha$ consists of all words of arbitrary length where each letter belongs to a G_{α_i} and consecutive letters are from different G_{α_i} .

Now suppose that X is decomposed as a union of path-connected spaces $\{A_\alpha\}$ each of which contains the base point $x_0 \in X$. Then inclusion $A_\alpha \hookrightarrow X \implies j_\alpha : \pi_1(A_\alpha) \rightarrow \pi_1(X)$ induce $\Phi : \star_\alpha \pi_1(A_\alpha) \rightarrow \pi_1(X)$.

Let $(\iota_{\alpha\beta})_* : \pi_1(A_\alpha \cap A_\beta) \rightarrow \pi_1(A_\alpha)$ be induced from inclusion. Hence,

$$(j_\alpha)_* \circ (\iota_{\alpha\beta})_* = (j_\beta)_* \circ (\iota_{\beta\alpha})_* : \pi_1(A_\alpha \cap A_\beta) \rightarrow \pi_1(X)$$

If $w \in \pi_1(A_\alpha \cap A_\beta)$ then $(\iota_{\alpha\beta})_*(w) ((\iota_{\beta\alpha})_*(w))^{-1}$ is in the kernel of Φ .

Let N denote the normal subgroup of $\star_\alpha \pi_1(A_\alpha)$ generated by such elements. Then $\ker(\Phi) = N$ (under assumptions).

Theorem 8.8 (Seifert-Van Kampen Theorem). If X is a union of path-connected open sets A_α such that $x_0 \in X$, and $x_0 \in A_\alpha$ for all α , and $A_\alpha \cap A_\beta$ is path-connected, and $A_\alpha \cap A_\beta \cap A_\gamma$ is also path-connected for all α, β, γ . Then Φ induces an isomorphism,

$$\pi_1(X) \cong \star_\alpha \pi_1(A_\alpha) / N$$

Proof. We have shown before that Φ is surjective. Now we will show that $\ker(\Phi) = N$. Define a factorization of an element $[f] \in \pi_1(X)$ as the product $[f_1] \cdots [f_k]$ such that each $[f_i] \in \pi_1(A_\alpha)$ for some A_α and $f \sim_p f_1 \star f_2 \star \cdots \star f_k$. A factorization of f is thus a word in $\star_\alpha \pi_1(A_\alpha)$. We will say that any two such factorizations are equivalent if they are related by the following two moves or their inverses:

- (1) Combine adjacent terms $[f_i][f_{i+1}]$ into a single term if $[f_i][f_{i+1}]$ are in the same group.
- (2) If $f_i \in A_\alpha \cap A_\beta$ we may regard $[f_i] \in \pi_1(A_\alpha)$ as $[f_i] \in \pi_1(A_\beta)$.

The first move does not change the factorization. The second move does not the element in $\star_\alpha \pi_1(A_\alpha)/N$.

Let us assume that $[f_1] \dots [f_k]$ and $[f'_1] \dots [f'_l]$ are two factorizations of $[f] \in \pi_1(X)$. We will show that they are equivalent. We have,

$$f_1 \star \dots \star f_k \cong f'_1 \star \dots \star f'_l$$

That is, there exists a path-homotopy $F : I \times I \rightarrow X$. Now choose a partition of $I \times I$ by

$$0 = s_0 < \dots < s_m = 1 \text{ and } 0 = t_0 < \dots < t_n = 1$$

such that each rectangle $R_{ij} = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ is mapped to a single A_α by F . We label this A_α by A_{ij} .

Now we will modify the partition in two ways:

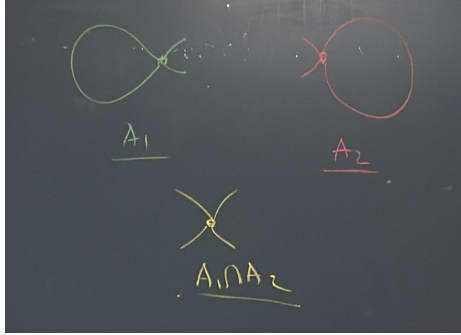
- (1) Add subdivisions to the top ($[t_{n-1}, t_n]$) and bottom row ($[t_0 - t_1]$) such that for each end point in the partition there exists a corresponding s_i .
- (2) Have at least 3 rows.
- (3) Every point in $I \times I$ must lie in at most 3 (smaller) rectangles
- (4) Rename the smaller rectangles (left to right, bottom to top) using R_i .

Define γ_r as the path from the left to the right edge of $I \times I$, and it separates rectangles $R_1, \dots, R_r$ from the rest of the rectangles.

Note that F_{γ_r} is a loop based at x_0 . And all F_{γ_r} are path homotopic. Then $F|_{\gamma_r}$ has a factorization. If $F(v) \neq x_0$, then we have a path from $x_0 \rightarrow F(v) = F|_{\gamma_1}$ $\square$

Example 8.2.

- (1) $\pi_1(\infty, x_0)$ (∞ denotes the figure 8 space).



Then, $\pi_1(\infty, x_0) \cong (\pi(A_1) \times \pi_1(A_2))/N = (\mathbb{Z} \times \mathbb{Z})/N$. Since $N = A_1 \cap A_2$ deformation retracts to a point this is $\mathbb{Z} \times \mathbb{Z}$.

- (2) $\pi_1(S^1 \vee S^2) \cong \mathbb{Z}$

$X_1 \wedge X_2 := (X_1 \sqcup X_2)/(x_1 \sim x_2)$ for some $x_1 \in X_1, x_2 \in X_2$.

- (3) $\pi_1(\mathbb{R}^3 \setminus S^1)$ **TODO:**

9. CELL COMPLEXES

Definition 9.1. We start with a discrete space whose points are regarded as 0-cells. Now, we inductively define the n -skeleton X^n from X^{n-1} by attaching n -cells e_α^n via maps $\phi_\alpha : S^{n-1} \rightarrow X^{n-1}$ where e_α^n s are D_α^n . Hence, $X^n = X^{n-1} \sqcup D_\alpha^n / (x \sim \phi_\alpha(x) \text{ if } \alpha, \forall x \in S^{n-1})$. We say a topological space X has a CW/CX structure if $X = \bigcup_n X^n$.

Definition 9.2. Each cell e_α^n of a CW CX has a map $\Phi_\alpha : D_\alpha^n \rightarrow X$ which extends $\phi_\alpha : \partial D_\alpha^n \rightarrow X$. It is defined by

$$D_\alpha^n \hookrightarrow X^{n-1} \sqcup D_\alpha^n \xrightarrow{\phi_\alpha : \partial D_\alpha^n \rightarrow X^{n-1}} X^{n-1} \sqcup D_\alpha^n = X^n \hookrightarrow X$$

This map is called the **characteristic map** associated to ϕ_α .

Definition 9.3. A **subcomplex** A of a CW CX X is a subspace $A \subset X$ such that A is a union of some number of cells in X with a attaching map for each of those cells has a image contained in A . We say that (X, A) is a CW pair.

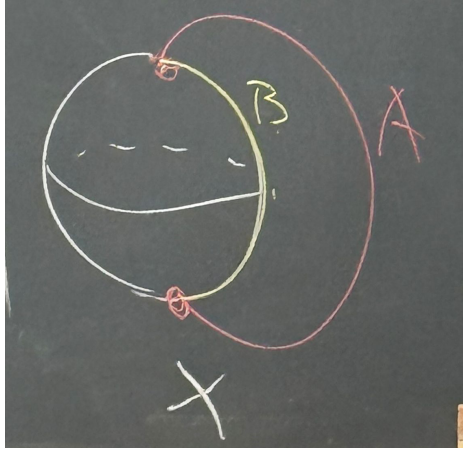
Definition 9.4 (Quotient CX). If (X, A) is a CW pair, then X/A inherits a canonical cell structure from X . The cells of X/A are the cells $X - A$, plus one 0-cell given by the image of A in X/A . For a cell e_α^n of $X - A$ with $\phi_\alpha : S^{n-1} \rightarrow X^{n-1}$, the corresponding attaching map in X/A is

$$S^{n-1} \rightarrow X^{n-1} \rightarrow X^{n-1}/A^{n-1}$$

Theorem 9.5. If (X, A) is a CW pair and A is contractible then $q : X \rightarrow X/A$ is a homotopy equivalence.

Proof. Hatcher Chapter 0, Homotopy Extension Lemma □

Example 9.1. (1) Consider a space X built from S^2 by attaching an arc to two points in S^2 .



Both (X, A) and (X, B) are CW complexes.

(2) $\pi_1(S^3, S^1) = ?$ **TODO:**

10. HOMOLOGY

Remark 10.1. $\pi_n(S^n) = \mathbb{Z}$, and $\pi_m(S^n) = 0$ if $m < n$. We don't know how to compute $\pi_m(S^n)$ for $m > n$.

Homology distinguishes higher dimension space and it is easy to compute. We define simplicial and singular homology separately, but eventually they have the same properties.

Remark 10.2. For spheres the homology groups $H_i(S^n)$ are isomorphic to the homotopy groups $\pi_i(S^n)$ for $1 \leq i \leq n$; and for $i > n$, $H_i(S^n) = 0$.

10.1. Simplicial Homology.

Definition 10.3. A *n -simplex* is the smallest convex set in $\mathbb{R}^m$ spanned by $(n + 1)$ points $v_0, \dots, v_n$ such that all of them do not lie in any hyperplane of dimension less than n . That is, $v_1 - v_0, v_2 - v_0, \dots, v_n - v_0$ are linearly independent. We say $(v_0, \dots, v_n)$ comes with the order $v_0, \dots, v_n$.

The standard n -simplex is

$$\left\{ (t_0, \dots, t_n) \in \mathbb{R}^{n+1} \mid \sum_i t_i = 1, t_i \geq 0 \right\}$$

There is a canonical homeomorphism from Δ^n to any n -simplex.

$$(t_0, \dots, t_n) \mapsto \sum t_i v_i$$

Definition 10.4. A *face* of an n -simplex $\Delta^n = [v_0, \dots, v_n]$ is the $n - 1$ -simplex obtained by deleting one of its $n + 1$ vertices. The vertices of a face, or of any subsimplex spanned by a subset of the vertices, will always be ordered according to their order in the larger simplex.

Definition 10.5. The *boundary* $\partial\Delta^n$ is the union of all the faces of Δ^n .

Definition 10.6. The *open simplex* $\mathring{\Delta}^n$ is the interior of Δ^n , i.e. $\Delta^n - \partial\Delta^n$.

Definition 10.7. The *Δ -complex structure* of a space X is a collection of maps $\sigma_\alpha : \Delta^n \rightarrow X$.

- (1) $\sigma_\alpha|_{\mathring{\Delta}^n}$ is injective and each point of X is in the image of exactly one such $\sigma_\alpha|_{\mathring{\Delta}^n}$.
- (2) Restriction of σ_α to a face of Δ^n is one of the maps $\sigma_\beta : \Delta^{n-1} \rightarrow X$. We identify the face of Δ^n with Δ^{n-1} by the canonical linear homeomorphism between them that preserves the ordering of the vertices.

$$\lambda_i : \Delta^{n-1} \rightarrow \Delta^n, \quad (t_0, \dots, t_{n-1}) \mapsto (t_0, \dots, t_{i-1}, 0, t_i, \dots, t_{n-1})$$

- (3) A set $U \subset X$ is open iff $\sigma_\alpha^{-1}(U)$ is open in Δ^n , for all α .

Definition 10.8. Let X be a Δ -complex and let $\Delta_n(X)$ denote the free abelian group with basis n -simplices of X . The elements of $\Delta_n(X)$ are of the form $\sum_i n_i \sigma_i$ where this is a finite sum, $n_i \in \mathbb{Z}$ and $\sigma_i : \Delta^n \rightarrow X$.

We denote the deletion of v_i from the simplex as $[v_0, \dots, \widehat{v_i}, \dots, v_n] \cong [v_0, \dots, v_{n-1}]$. We will always orient the 1-simplex $[v_i, v_j]$ by an arrow going from v_i to v_j for $i < j$.

Definition 10.9. We define a **boundary homomorphism** $\partial_n : \Delta_n(X) \rightarrow \Delta_{n-1}(X)$ where,

$$\partial_n(\sigma_\alpha) = \sum_i (-1)^i \sigma_\alpha|_{[v_0, \dots, \widehat{v}_i, \dots, v_n]}$$

Proposition 10.10. The composition

$$\Delta_n(X) \xrightarrow{\partial_n} \Delta_{n-1}(X) \xrightarrow{\partial_{n-1}} \Delta_{n-2}(X)$$

is zero ($\partial_{n-1} \circ \partial_n = 0$).

Proof.

$$\begin{aligned} \partial_n(\sigma) &= \sum_i (-1)^i \sigma|_{[v_0, \dots, \widehat{v}_i, \dots, v_n]} \\ \partial_{n-1} \circ \partial_n(\sigma) &= \sum_{j < i} (-1)^i (-1)^j \sigma|_{[v_0, \dots, \widehat{v}_j, \dots, \widehat{v}_i, \dots, v_n]} + \sum_{i < j} (-1)^i (-1)^{j-1} \sigma|_{[v_0, \dots, \widehat{v}_i, \dots, \widehat{v}_j, \dots, v_n]} \end{aligned}$$

□

Definition 10.11. In general, whenever we have (C_n, ∂_n)

$$\cdots \rightarrow C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}} \cdots$$

with $\partial_n \circ \partial_{n+1} = 0$, then $\text{img}(\partial_{n+1}) \subset \ker(\partial_n)$. Hence we define the ***n*-th homology group**,

$$H_n := \ker(\partial_n) / \text{im}(\partial_{n+1})$$

where elements of $\ker(\partial_n)$ are ***cycles*** and elements of $\text{im}(\partial_{n+1})$ are ***boundaries***. The ***n*-th simplicial homology** of X is $H_n^\Delta(X) = \ker(\partial_n) / \text{im}(\partial_{n+1})$.

Example 10.1. Let $X = S^1$. There is one 0-simplex v and one 1-simplex e . Then,

$$\Delta_0(S^1) = \mathbb{Z}\langle v \rangle, \Delta_1(S^1) = \mathbb{Z}\langle e \rangle, \Delta_2(S^1) = 0$$

And we have,

$$0 \xrightarrow{\partial_2} \Delta_1(S^1) \xrightarrow{\partial_1} \Delta_0(S^1) \xrightarrow{\partial_0} 0$$

with

$$H_0^\Delta(S^1) = \frac{\ker(\partial_0)}{\text{im}(\partial_1)} = \frac{\mathbb{Z}\langle v \rangle}{\{\sigma_1|_{[v_1]} - \sigma_1|_{[v_0]}\}} = \frac{\mathbb{Z}\langle v \rangle}{\{0\}} \cong \mathbb{Z}\langle v \rangle \cong \mathbb{Z},$$

$$H_1^\Delta(S^1) = \frac{\ker(\partial_1)}{\text{im}(\partial_2)} = \frac{\mathbb{Z}\langle v \rangle}{\{0\}} \cong \mathbb{Z}\langle v \rangle \cong \mathbb{Z},$$

and

$$H_2^\Delta(S^1) = \frac{\ker(\partial_2)}{\text{im}(\partial_3)} = 0$$

We conclude that $H_n^\Delta(S^1) \cong \mathbb{Z}$ if $n = 0, 1$ and $H_n^\Delta(S^1) = 0$ otherwise.

Example 10.2.

$$\begin{array}{ccc}
 v_0 & \xrightarrow{a} & v_0 \\
 b \uparrow & \nearrow L & \uparrow b \\
 & c & \\
 v_0 & \xrightarrow{a} & v_0 \\
 & U &
 \end{array}$$

We have,

- $\Delta_0(T^2) = \mathbb{Z}\langle v_0 \rangle$
- $\Delta_1(T^2) = \mathbb{Z}\langle a \rangle \oplus \mathbb{Z}\langle b \rangle \oplus \mathbb{Z}\langle c \rangle$
- $\Delta_2(T^2) = \mathbb{Z}\langle U \rangle \oplus \mathbb{Z}\langle L \rangle$
- $\Delta_3(T^2) = 0$

$$0 \xrightarrow{\partial_3} \Delta_2(T^2) \xrightarrow{\partial_2} \Delta_1(T^2) \xrightarrow{\partial_1} \Delta_0(T^2) \xrightarrow{\partial_0} 0$$

The homologies are,

- $H_0^\Delta(T^2) = \frac{\ker(\partial_0)}{\text{im}(\partial_1)} = \frac{\mathbb{Z}\langle v_0 \rangle}{\{0\}} \cong \mathbb{Z}\langle v_0 \rangle \cong \mathbb{Z}$
- $H_1^\Delta(T^2) = \frac{\ker(\partial_1)}{\text{im}(\partial_2)} = \frac{\mathbb{Z}\langle a \rangle \oplus \mathbb{Z}\langle b \rangle \oplus \mathbb{Z}\langle c \rangle}{\mathbb{Z}\langle a+b-c \rangle} \cong \frac{\langle a, b, c \rangle}{\langle a+b-c \rangle} \cong \langle a, b \rangle \cong \mathbb{Z}\langle a \rangle \oplus \mathbb{Z}\langle b \rangle$
- $H_2^\Delta(T^2) = \frac{\ker(\partial_2)}{\text{im}(\partial_3)} = \frac{\mathbb{Z}\langle U-L \rangle}{0} \cong \mathbb{Z}\langle U-L \rangle \cong \mathbb{Z}$
- $H_n^\Delta = 0$ if $n \geq 3$.

Example 10.3. Compute $H_n^\Delta(\mathbb{RP}^2)$.

$$\begin{array}{ccc}
 v_0 & \xleftarrow{b} & v_1 \\
 a \uparrow & \nearrow U & \downarrow a \\
 & c & \\
 v_1 & \xrightarrow{b} & v_0 \\
 & L &
 \end{array}$$

We have,

- $\Delta_0(\mathbb{RP}^2) = \mathbb{Z}\langle v_0 \rangle \oplus \mathbb{Z}\langle v_1 \rangle$
- $\Delta_1(\mathbb{RP}^2) = \mathbb{Z}\langle a \rangle \oplus \mathbb{Z}\langle b \rangle \oplus \mathbb{Z}\langle c \rangle$
- $\Delta_2(\mathbb{RP}^2) = \mathbb{Z}\langle U \rangle \oplus \mathbb{Z}\langle L \rangle$

$$0 \xrightarrow{\partial_3} \Delta_2(\mathbb{RP}^2) \xrightarrow{\partial_2} \Delta_1(\mathbb{RP}^2) \xrightarrow{\partial_1} \Delta_0(\mathbb{RP}^2) \xrightarrow{\partial_0} 0$$

The homologies are,

- $H_0^\Delta(\mathbb{RP}^2) = \frac{\ker(\partial_0)}{\text{im}(\partial_1)} = \frac{\mathbb{Z}\langle v_0 \rangle \oplus \mathbb{Z}\langle v_1 \rangle}{\langle v_0 - v_1 \rangle} \cong \mathbb{Z}\langle v_0 \rangle \cong \mathbb{Z}$
- $H_1^\Delta(\mathbb{RP}^2) = \frac{\ker(\partial_1)}{\text{im}(\partial_2)} = \frac{\mathbb{Z}\langle a-b \rangle \oplus \mathbb{Z}\langle c \rangle}{\mathbb{Z}\langle a-b-c, a-b+c \rangle}$

Let $x = a - b$, $y = c$. Then we have,

$$x + y = 0, \quad x - y = 0 \implies 2x = 2y = 0$$

i.e. both x and y have order 2. And,

$$x - y = 0 \implies x = y$$

So,

$$H_1^\Delta(\mathbb{RP}^2) = \frac{\mathbb{Z}\langle a-b \rangle \oplus \mathbb{Z}\langle c \rangle}{\mathbb{Z}\langle a-b-c, a-b+c \rangle} \cong \frac{\mathbb{Z}\langle x \rangle}{2x} \cong \mathbb{Z}/2\mathbb{Z}$$

- $H_2^\Delta(\mathbb{RP}^2) = \frac{\ker(\partial_2)}{\text{im}(\partial_3)} = \frac{\{0\}}{\{0\}} \cong \{0\}$
- $H_n^\Delta = 0$ if $n \geq 3$.

Definition 10.12. A **singular n -simplex** in a space X is a map $\sigma : \Delta^n \rightarrow X$, where there are no conditions of $\text{im}(\sigma)$ restricted to a face (i.e. singularities are allowed). The only requirement is that σ is continuous.

Definition 10.13. A **chain complex** $C_n(X)$ is the free abelian group with the set of singular n -simplices in X as basis. Its elements are called **n -chains** which are (finite) formal sum of singular n -simplices.

$$C_n(X) := \left\{ \sum_i n_i \sigma_i \mid \sigma_i : \Delta^n \rightarrow X \right\}$$

with the differential $\partial_n : C_n(X) \rightarrow C_{n-1}(X)$ where,

$$\partial_n(\sigma) := \sum_i (-1)^i \sigma|_{[v_0, \dots, \widehat{v_i}, \dots, v_n]}$$

and $\partial_n \circ \partial_{n+1} = 0$. The **n -th singular homology** is

$$H_n(X) := \frac{\ker(\partial_n)}{\text{im}(\partial_{n+1})}$$

Remark 10.14. $H_n(X)$ is an invariant under the homeomorphism class of X .

Proposition 10.15. Let $X = \sqcup X_\alpha$ then $H_n(X) \cong \oplus_\alpha H_n(X_\alpha)$, where each X_α is a distinct path connected component of X .

Proof. We have $C_n(X) \cong \oplus_\alpha C_n(X_\alpha)$ and $\partial_n = \oplus_\alpha \partial_n^\alpha$ which implies

$$H_n(X) \cong \oplus_\alpha H_n(X_\alpha)$$

□

Proposition 10.16. If X is non-empty and path-connected then $H_0(X) \cong \mathbb{Z}$.

Proof. By definition,

$$H_0(X) \cong \frac{C_0(X)}{\text{im}(\partial_1)}$$

We define a homomorphism

$$\varepsilon : C_0(X) \rightarrow \mathbb{Z}, \quad \sum_i n_i \sigma_i \mapsto \sum_i n_i$$

This map is surjective since for any $n \in \mathbb{Z}$, $\varepsilon(n\sigma_0) = n$ where $\sigma_0 : \Delta^0 \rightarrow X$.

Claim: $\ker(\varepsilon) = \text{im}(\partial_1)$.

Proof. Note that $\text{im}(\partial_1) \subseteq \ker(\varepsilon)$ since for a singular 1-simplex $\sigma : \Delta^1 \rightarrow X$ we have $\varepsilon(\partial_1(\sigma)) = 0$.

Now for $\ker(\varepsilon) \subseteq \text{im}(\partial_1)$, let $\varepsilon(\sum_i n_i \sigma_i) = 0 \implies \sum_i n_i = 0$.

Choose paths $\tau : I \rightarrow X$ for $i > 0$, from a base point $x_0 \in X$ to $\sigma_i(v_0)$ and define $\sigma_0(v_0) = x_0$. We can view $\tau_i : [v_0, v_i] \rightarrow X$ as a singular 1-simplex. Thus, $\delta(\tau_i) = \sigma_i(v_0) - x_0$. Hence,

$$\partial \left(\sum_i n_i \tau_i \right) = \sum_i n_i \sigma_i(v_0) - \sum_i n_i \cdot x_0 = \sum_i n_i \sigma_i(v_0) \implies \sum_i n_i \sigma_i \in \text{im}(\partial_i)$$

Thus $\ker(\varepsilon) = \text{im}(\partial_1)$. □

That is,

$$H_0(X) \cong \frac{C_0(X)}{\text{im}(\partial_1)} = \frac{C_0(X)}{\ker(\varepsilon)} \cong \mathbb{Z}$$

□

Proposition 10.17. *If X is a point, then $H_n(X) \cong 0$ for $n > 0$ and $H_0(X) \cong \mathbb{Z}$.*

Proof. Let x_0 be a point. Then the constant map $\sigma_n : \Delta^n \rightarrow X$ gives us a n -simplex for all n since constant maps are continuous. Since the restriction of the constant map is the constant map we get,

$$\partial_n(\sigma_n) = \sum_{i=0}^n (-1)^i \sigma_{n-1} = \sigma_{n-1} \sum_{i=0}^n (-1)^i = \begin{cases} 0, & n \text{ odd} \\ \sigma_{n-1}, & n \text{ even} \end{cases}$$

Thus we get,

$$\begin{aligned} \dots &\xrightarrow{\partial_3} C_2(x_0) \xrightarrow{\partial_2} C_1(x_0) \xrightarrow{\partial_1} C_0(x_0) \xrightarrow{\partial_0} 0 \\ &\dots \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0 \end{aligned}$$

Then for odd n we have, $\ker(\partial_n) = \mathbb{Z}$ and $\text{im}(\partial_{n+1}) = \text{im}(\text{id}_{C_{n-1}}) = \mathbb{Z}$, so

$$H_n(x_0) = \frac{\ker(\partial_n)}{\text{im}(\partial_{n+1})} = \frac{\mathbb{Z}}{\mathbb{Z}} = \{0\}$$

For even $n > 0$, we have $\ker(\partial_n) = 0$, so

$$H_n(x_0) = \frac{\ker(\partial_n)}{\text{im}(\partial_{n+1})} = \{0\}$$

For $n = 0$, we have,

$$H_0(x_0) = \frac{\ker(\partial_0)}{\text{im}(\partial_1)} = \frac{\mathbb{Z}}{0} = \mathbb{Z}$$

Hence, the singular homology groups of a point are 0 for $n > 0$ and $\mathbb{Z}$ for $n = 0$. □

Definition 10.18. We will change the definition of homology so that a point has trivial $H_0(-)$. The **reduced homology groups** $\widetilde{H}_n(X)$ is defined to be the homology of the following chain complex.

$$\rightarrow C_n(X) \xrightarrow{\partial_n} C_{n-1}(X) \xrightarrow{\partial_{n-1}} \dots \xrightarrow{\partial_1} C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \xrightarrow{0} 0$$

So, $\widetilde{H}_n(X) \cong H_n(X)$ if $n \geq 1$. Moreover we have $\text{im}(\partial_1) \subseteq \ker(\varepsilon)$. So,

$$H_0(X) = \frac{C_0(X)}{\text{im}(\partial_1)} \cong \frac{\ker(\varepsilon) \oplus \mathbb{Z}}{\text{im}(\partial_1)} = \widetilde{H}_0(X) \oplus \mathbb{Z}$$

Hence, $\widetilde{H}_0(pt) = 0$.

Definition 10.19. If $\alpha \in C_n(X)$ we say that α is a **chain**. α is a **cycle** if $\partial_n(\alpha) = 0$, and α is a **boundry** if $\alpha = \partial(\beta)$ for some $\beta \in C_{n+1}(X)$.

Definition 10.20. A map $\phi : C_n(X) \rightarrow C_n(Y)$ is called a **chain map** if $\partial(\phi) = \phi(\partial)$. Such a ϕ always induces a map in homology.

Proposition 10.21 (Homotopy Invariance of Singular Homology). Let $f : X \rightarrow Y$ then $\exists$ an induced homomorphism defined by

$$f_{\#} : C_n(X) \rightarrow C_n(Y), \quad \sigma \mapsto f \circ \sigma$$

which extends $\mathbb{Z}$ -linearly. This map satisfies $f_{\#}\partial = \partial f_{\#}$.

Proof. For some $\sigma \in C_n(X)$.

$$\begin{aligned} f_{\#}\partial(\sigma) &= f_{\#} \left(\sum_i (-1)^i \sigma | [v_0, \dots, \widehat{v}_i, \dots, v_n] \right) \\ &= \sum_i (-1)^i (f \circ \sigma) |_{[v_0, \dots, \widehat{v}_i, \dots, v_n]} \\ &= \partial f_{\#}(\sigma) \end{aligned}$$

Hence $f_{\#}$ induces a map $f_{\star} : H_n(X) \rightarrow H_n(Y)$.

Proof. Note $f_{\#}(\ker \partial_n) \subset \ker \partial_n$. Since $\alpha \in C_n(X)$ such that $\partial(\alpha) = 0$. Then,

$$f_{\#}(\partial_X(\alpha)) = 0 = \partial(f_{\#}(\alpha)) \implies f_{\#}(\alpha) \in \ker(\partial_n)$$

Again note that $f_{\#}(\text{im } \partial_{n+1}) \subset \text{im } \partial_{n+1}$. Let $\alpha \in C_n(X)$ and $\beta \in C_{n+1}(X)$ such that $\alpha = \partial(\beta)$. Then,

$$f_{\#}(\alpha) = 0 = f_{\#}\partial(\beta) = \partial(f_{\#}\beta) \implies f_{\#}(\alpha) \in \text{im}(\partial_{n+1})$$

Since $f_{\#}$ sends $\ker(\partial_n)$ to $\ker(\partial_n)$ and $\text{im}(\partial_{n+1})$ to $\text{im}(\partial_{n+1})$ it induces a map $f_{\star} : H_n(X) \rightarrow H_n(Y)$. □

□

Theorem 10.22. If $f, g : X \rightarrow Y$ are homotopic, then they induce the same map on homology,

$$f_{\star} = g_{\star} : H_n(X) \rightarrow H_n(Y)$$

Proof. The idea is to subdivide $\Delta^n \times I$ into $(n+1)$ -simplices. Let's say $\Delta^n \times \{0\} = [v_0, \dots, v_n]$, $\Delta^n \times \{1\} = [w_0, \dots, w_n]$. We start with $[v_0, \dots, v_n]$ and consider the $(n+1)$ -simplex $[v_0, v_n, w_n]$. Here we keep track of a distinguished edge from $v_n \rightarrow w_n$. Note that by contracting along the distinguished edge we go from the $[v_0, \dots, v_n]$ -face to $[v_0, \dots, v_{n-1}, w_n]$ -face of the $(n+1)$ -simplex. We say that $[v_0, \dots, v_n]$ is the lower face and $[v_0, \dots, v_{n-1}, w_n]$ the upper face.

In a typical step, we will construct a $(n+1)$ -simplex $[v_0, \dots, v_i, w_i, \dots, w_n]$ with lower face $[v_0, \dots, v_i, w_{i+1}, \dots, w_n]$ and upper face $[v_0, \dots, v_{i-1}, w_i, \dots, w_n]$ which we get by contracting along the edge from v_i to w_i . $\square$

Proposition 10.23. Now given $F : X \times I \rightarrow Y$ a homotopy we can define for any singular n -simplex $\sigma : \Delta^n \rightarrow X$ the map,

$$F' : F \circ (\sigma \times \text{Id}) : \Delta^n \times I \rightarrow Y$$

Now define the **prism operator** $P : C_n(X) \rightarrow C_{n+1}(Y)$ by

$$P(\sigma) = \sum_i (-1)^i F' |_{[v_0, \dots, v_i, w_i, \dots, w_n]}$$

It will turn out that P is not a chain map but

$$\partial P = g_{\#} - f_{\#} - P\partial$$

Proof. Let $\sigma : \Delta^n \rightarrow X$. Then,

$$\partial P(\sigma) = \sum_{j \leq i} (-1)^i (-1)^j F' |_{[v_0, \dots, \widehat{v}_j, \dots, v_i, w_i, \dots, w_n]} + \sum_{j \geq i} (-1)^i (-1)^{j+1} F' |_{[v_0, \dots, v_i, w_i, \dots, \widehat{v}_j, \dots, w_n]}$$

all terms of $\partial P(\sigma)$ with $i = j$ will cancel except $F' |_{[\widehat{v}_0, w_0, \dots, w_n]} = g \circ \sigma = g_{\#}(\sigma)$ and $-F' |_{[v_0, \dots, v_n, \widehat{w}_n]} = -f \circ \sigma = -f_{\#}(\sigma)$ that is $\partial P(\sigma) = g_{\#}\sigma - f_{\#}\sigma$ which is a homotopy between $g_{\#}$ and $f_{\#}$. The terms with $i \neq j$ are exactly $-P\partial(\sigma)$, that is,

$$P\partial(\sigma) = \sum_{j < i} (-1)^{i-1} (-1)^j F' |_{[v_0, \dots, \widehat{v}_j, \dots, v_i, w_i, \dots, w_n]} + \sum_{j > i} (-1)^i (-1)^j F' |_{[v_0, \dots, v_i, w_i, \dots, \widehat{v}_j, \dots, w_n]}$$

Thus,

$$\partial P = g_{\#} - f_{\#} - P\partial$$

$\square$

Corollary 10.23.1. $\widetilde{H}_i(D^n) = 0$ for all i .

11. EXACT SEQUENCES

Theorem 11.1. Let X be a space and A be a subspace of X such that A is closed and it is a deformation retraction of a neighborhood of A in X . Then we have a long exact sequence given by

$$\dots \rightarrow \widetilde{H}_n(A) \xrightarrow{i_*} \widetilde{H}_n(X) \xrightarrow{j_*} \widetilde{H}_n(X/A) \xrightarrow{\partial} \widetilde{H}_{n-1}(A) \xrightarrow{i_*} \widetilde{H}_{n-1}(X) \xrightarrow{j_*} \widetilde{H}_{n-1}(X/A) \xrightarrow{\partial} \dots$$

where i_* is induced from $i : A \hookrightarrow X$ and j_* is induced from $j : X \rightarrow X/A$.

Definition 11.2. If (X, A) satisfies the hypotheses of the previous definition of LES, say (X, A) is “good”.

Remark 11.3. Any CW pair (complex, subcomplex) (X, A) is good, i.e. admits a LES. (Hatcher for proof.)

Corollary 11.3.1. $\widetilde{H}_i(S^n) \cong \mathbb{Z}$ if $i = n$, and $= 0$ otherwise.

Proof. Consider the LES of the pair (D^n, S^{n-1}) . Since $D^n/S^{n-1} \cong S^n$, the LES becomes

$$\widetilde{H}_i(S^{n-1}) \rightarrow \widetilde{H}_i(D^n) \xrightarrow{0} \widetilde{H}_i(S^n) \rightarrow \widetilde{H}_{i-1}(S^{n-1}) \rightarrow \widetilde{H}_{i-1}(D^n)$$

Since D^n is contractible, $\widetilde{H}_i(D^n) = 0$ for all i , so:

$$0 \rightarrow \widetilde{H}_i(S^n) \xrightarrow{\partial} \widetilde{H}_{i-1}(S^{n-1}) \rightarrow 0$$

Thus $H_i(S^n) \cong H_{i-1}(S^{n-1})$ for all i , and the computation follows by induction. $\square$

Example 11.1. $H_0(S^0) \cong \mathbb{Z} \oplus \mathbb{Z}$, since S^0 has 2 path-connected components, so $\widetilde{H}_0(S^0) = \mathbb{Z}$.

Corollary 11.3.2. S^n is not homeomorphic to S^m when $n \neq m$.

Definition 11.4. Given a space X and a subspace A (where (X, A) is not necessarily good), we define

$$C_n(X, A) := C_n(X) / C_n(A).$$

Note $\partial : C_n(X) \rightarrow C_{n-1}(X)$ sends $C_n(A) \rightarrow C_{n-1}(A)$, hence induces a map $\partial : C_n(X, A) \rightarrow C_{n-1}(X, A)$.

We get a sequence

$$\dots \rightarrow C_n(X, A) \rightarrow C_{n-1}(X, A) \rightarrow C_{n-2}(X, A) \rightarrow \dots$$

where $\partial^2 = 0$. We denote the relative homology by

$$H_n(X, A) := \ker \partial / \text{im } \partial.$$

Elements of $H_n(X, A)$ are relative cycles, equivalence classes of $\alpha \in C_n(X)$ such that $\partial\alpha \in C_{n-1}(A)$ (i.e. “ $\partial\alpha = 0$ ”). A relative cycle α is trivial if it is a relative boundary, i.e. $\alpha = \partial\beta + \gamma$ for some $\beta \in C_{n+1}(X)$, $\gamma \in C_n(A)$.

TODO:

Theorem 11.5 (Long Exact Sequence of a Pair).

$$\dots \rightarrow H_n(A) \rightarrow H_n(X) \rightarrow H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \rightarrow \dots$$

(Proof: homework/reading.)

This follows from the “SES of chain complexes gives rise to LES in homology”: given 3 chain complexes (A_n, ∂_n) , (B_n, ∂_n) , (C_n, ∂_n) (with different notation) and $0 \rightarrow C_n(A) \rightarrow C_n(B) \rightarrow C_n(C) \rightarrow 0$ exact for all n , we get

$$\begin{array}{ccccccc} 0 & \rightarrow & C_n(A) & \xrightarrow{i_*} & C_n(B) & \xrightarrow{j_*} & C_n(C) \rightarrow 0 \\ & & \downarrow \partial & & \downarrow \partial & & \downarrow \partial \\ 0 & \rightarrow & C_{n-1}(A) & \xrightarrow{i_*} & C_{n-1}(B) & \xrightarrow{j_*} & C_{n-1}(C) \rightarrow 0 \end{array}$$

with $\partial i_* = i_* \partial$ and $\partial j_* = j_* \partial$, and LES:

$$\cdots \rightarrow H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{j_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \xrightarrow{j_*} H_{n-1}(C) \xrightarrow{\partial} \cdots$$

In our case, the SES is $0 \rightarrow C_n(A) \xrightarrow{i_*} C_n(X) \xrightarrow{j_*} C_n(X)/C_n(A) \rightarrow 0$, given by inclusion on 8 quotient maps.

Description of ∂ : Given $[\alpha] \in H_n(X, A)$, then by definition $\partial\alpha \in C_{n-1}(A)$. Define $\partial[\alpha] = [\partial\alpha] \in H_{n-1}(A)$. Then $\partial(\partial\alpha) = 0$.

Example 11.2. $H_n(X, x_0) \cong \tilde{H}_n(X)$ for any $x_0 \in X$.

Proof. LES with $X = X$, $A = x_0$; since x_0 is reduced, $H_n(x_0) = 0$, so

$$\tilde{H}_n(X) \cong \tilde{H}_n(X, x_0).$$

Then $H_n(X, A) \cong \tilde{H}_n(X, A)$ implies $\tilde{H}_n(X) \cong H_n(X, x_0)$. □

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Goal: $H_n(X, A) \cong H_n(X', A')$ for some special $X' \subseteq X$, $A' \subseteq A$.

Theorem 11.6 (Excision). Let $A, B \subseteq X$ be subspaces whose interiors cover X . Then the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ induces an isomorphism

$$H_n(B, A \cap B) \cong H_n(X, A) \quad \text{for all } n \in \mathbb{N}.$$

Idea: One can define a chain complex $C_n(A + B)$ as the subgroup generated by chains in $C_n(X)$ which can be written as a sum of a chain in A and a chain in B . The idea is that $C_n(A + B) \hookrightarrow C_n(X)$ induces an isomorphism in homology. (See Hatcher for proof — “supposedly beautiful.”)

Theorem 11.7. If nonempty open sets $U \subseteq \mathbb{R}^n$, $V \subseteq \mathbb{R}^m$ are homeomorphic, then $n = m$.

Proof. For $x \in U$, consider $\mathbb{R}^n = U \cup \mathbb{R}^n \setminus \{x\}$. Then by excision,

$$H_k(U, U \setminus \{x\}) \cong H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \quad \forall k.$$

Compute $H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\})$ via the LES:

$$\rightarrow H_k(\mathbb{R}^n \setminus \{x\}) \rightarrow H_k(\mathbb{R}^n) \rightarrow H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \rightarrow H_{k-1}(\mathbb{R}^n \setminus \{x\}) \rightarrow H_{k-1}(\mathbb{R}^n)$$

Since $\mathbb{R}^n \setminus \{x\} \simeq S^{n-1}$ and $\mathbb{R}^n$ is contractible:

$$H_k(S^{n-1}) \rightarrow 0 \rightarrow H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \xrightarrow{\cong} H_k(S^{n-1}) \rightarrow 0$$

So $H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \cong H_k(S^{n-1}) \cong \mathbb{Z}$ if $k = n$, and 0 otherwise.

If $f : U \rightarrow V$ is a homeomorphism, then f induces $f_* : (U, U \setminus \{x\}) \xrightarrow{\text{homeo}} (V, V \setminus \{f(x)\})$, so

$$f_* : H_k(U, U \setminus \{x\}) \rightarrow H_k(V, V \setminus \{f(x)\})$$

is an isomorphism, giving $H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \cong H_k(\mathbb{R}^m, \mathbb{R}^m \setminus \{f(x)\})$.

The LHS is $\mathbb{Z}$ when $k = n$ and 0 otherwise; the RHS is $\mathbb{Z}$ when $k = m$ and 0 otherwise. Therefore $n = m$. $\square$

Equivalence of Simplicial and Singular Homology. Put a Δ -complex structure on X , then show $H_n^\Delta(X) \cong H_n(X)$.

To show this, we need the following definition.

Definition 11.8. If $A \subseteq X$ is a subspace, the relative simplicial chain complex is

$$\Delta_n(X, A) := \Delta_n(X) / \Delta_n(A),$$

with $\partial : \Delta_n(X) \rightarrow \Delta_{n-1}(X)$ inducing $\partial : \Delta_n(X, A) \rightarrow \Delta_{n-1}(X, A)$. Define

$$H_n^\Delta(X, A) := \ker \partial / \text{im } \partial.$$

We get a SES $0 \rightarrow \Delta_n(A) \rightarrow \Delta_n(X) \rightarrow \Delta_n(X, A) \rightarrow 0$ and hence a LES in homology $H_n^\Delta \quad (*)$.

Recall $\Delta_n(X) = \{\text{maps } \Delta^n \rightarrow X\}$ with some nice properties, and $C_n(X) = \{\text{maps } \Delta^n \rightarrow X\}$. There is a canonical map $\phi : \Delta_n(X) \rightarrow C_n(X)$, $(\Delta^n \rightarrow X) \mapsto (\Delta^n \rightarrow X)$, forgetting the nice properties.

Claim: ϕ induces an isomorphism $H_n^\Delta(X) \xrightarrow{\sim} H_n(X)$.

Today we prove this claim for the case where X is finite-dimensional, i.e. $X^k = \emptyset$ for all $k \geq k_0$ (X^k is the k -skeleton, using simplices of dimension $\leq k$).

Proof. We have a commutative diagram with LES rows:

$$\begin{array}{ccccccccc} \rightarrow & H_{n+1}^\Delta(X^k, X^{k-1}) & \rightarrow & H_n^\Delta(X^{k-1}) & \rightarrow & H_n^\Delta(X^k) & \rightarrow & H_n^\Delta(X^k, X^{k-1}) & \rightarrow & H_{n-1}^\Delta(X^{k-1}) & \rightarrow \\ & \downarrow \phi_* & & \downarrow \phi_* & & \downarrow \phi_* & & \downarrow \phi_* & & \downarrow \phi_* & \\ \rightarrow & H_{n+1}(X^k, X^{k-1}) & \rightarrow & H_n(X^{k-1}) & \rightarrow & H_n(X^k) & \rightarrow & H_n(X^k, X^{k-1}) & \rightarrow & H_{n-1}(X^{k-1}) & \rightarrow \end{array}$$

We show maps ① and ④ are isomorphisms.

$$\text{Note } \Delta_n(X^k, X^{k-1}) \cong \Delta_n(X^k) / \Delta_n(X^{k-1}) = \begin{cases} \Delta_n(X^k) & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}.$$

Now if $d \in \Delta_n(X^n)$, then $\partial d \in \Delta_{n-1}(X^{n-1})$ by face properties, so $\partial_{\text{rel}} = \Delta_n(X^n) \rightarrow 0$ implies $\text{Im } \partial_{\text{rel}} = 0$. Since $\partial(\cdot) = 0$, $H_n^\Delta(X^k, X^{k-1})$ is free abelian for $n = k$ and 0 otherwise.

Now consider $\bigsqcup_\alpha \Delta_\alpha^k / \bigsqcup_\alpha \partial \Delta_\alpha^k \xrightarrow{\sim} (X^k, X^{k-1})$.

But $\bigsqcup_\alpha \Delta_\alpha^k / \bigsqcup_\alpha \partial \Delta_\alpha^k \cong X^k / X^{k-1}$ by quotient map / homeomorphism ($\cong$ means homeomorphism here).

n -simplices (quotient by boundary): $\bigvee_{\alpha} S^k := \{\bigcup_{\alpha} S^k / x_{\alpha} \in S^k_{\alpha} \times \{\beta\}, \forall \alpha, \beta\}$ (wedge product), so:

$$H_n\left(\bigvee_{\alpha} S^k\right) \cong \bigoplus_{\alpha} H_n(S^k) \rightarrow \text{free abelian for } n = k, 0 \text{ otherwise.}$$

So the domains/codomains of ① and ④ in the diagram are isomorphic. Why, induced by ϕ , though? Because ϕ is almost the “identity” map; the isomorphism is the “same” map, so we use it.

By the Five Lemma (applied to maps ② and ③ isomorphisms), we get ② and ⑤ isomorphisms. The five lemma says: if you have an exact sequence

$$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E, \quad A \rightarrow A', B \rightarrow B', D \rightarrow D', E \rightarrow E' \text{ isos,}$$

then $C \rightarrow C'$ is also an isomorphism.

Repeat until $X = X^{k_0}$, and remaining things are 0. Done. $\square$

Used in proof:

Theorem 11.9. *If (X, A) is a good pair, then $q : (X, A) \rightarrow (X/A, A/A)$ induces an isomorphism*

$$q_* : H_n(X, A) \rightarrow H_n(X/A, A/A) \cong H_n(X/A, \{p\}) \cong \tilde{H}_n(X/A).$$

This gives $H_n^{\Delta}(X^k, X^{k-1}) \cong H_n^{\Delta}(X^k/X^{k-1})$. Then since $X^k/X^{k-1} \cong \bigvee S^k$, use the LES of $(\bigcup S^k, \bigcup \{x_{\alpha}\})$ and $\tilde{H}_n(\{p\}) = 0$:

$$\tilde{H}_n\left(\bigcup S^k, \bigcup x_{\alpha}\right) \cong \tilde{H}_n\left(\bigcup S^k\right) \cong \bigoplus_{\alpha} H_n(S^k).$$

Mayer–Vietoris Sequence. Suppose $X = A \cup B$, with $C_n(A+B) \hookrightarrow C_n(X)$.

Claim: there is a SES

$$0 \rightarrow C_n(A \cap B) \xrightarrow{\phi} C_n(A) \oplus C_n(B) \xrightarrow{\psi} C_n(A+B) \rightarrow 0$$

where $\phi(x) = (x, -x)$ and $\psi(x, y) = x + y$.

If we prove it's exact, we'll get a LES in homology.

- ϕ is injective since $\phi(x) = 0 \Rightarrow x = 0$.
- $\ker \psi = \{(x, y) \mid x + y = 0\} = \{(x, -x) \mid x \in C_n(A), -x \in C_n(B)\} = \{(x, -x) \mid x \in C_n(A \cap B)\} = \text{im } \phi$.
- ψ is surjective (from definition of $C_n(A+B)$).

Thus the sequence is exact, so we get a LES in homology, called the **Mayer–Vietoris sequence**:

$$\cdots \rightarrow H_n(A \cap B) \rightarrow H_n(A) \oplus H_n(B) \xrightarrow{\psi_*} H_n(A+B) \xrightarrow{\partial} H_{n-1}(A \cap B) \rightarrow \cdots$$

(also has a reduced version.)

Note: If $A \cap B$ is path-connected, then

$$\tilde{H}_1(A \cap B) \xrightarrow{\phi_*} \tilde{H}_1(A) \oplus \tilde{H}_1(B) \xrightarrow{\psi_*} \tilde{H}_1(X) \rightarrow \tilde{H}_0(A \cap B)^0$$

then $H_1(X) \cong H_1(A) \oplus H_1(B) / \ker \psi_* \cong H_1(A) \oplus H_1(B) / \operatorname{im} \phi_*$. (Similar to Van Kampen's theorem.)

Remark 11.10. H_1 is the abelianization of π_1 . In particular, $H_1(\Sigma_g) \cong \pi_1(\Sigma_g)^{\operatorname{ab}} \cong \mathbb{Z}^{2g}$.

Suspension of a Space. Given a space X , its *suspension* is

$$\Sigma X := X \times [-1, 1] / (x_1 \sim x_2, x \times \{1\} \sim x \times \{-1\} \text{ for } x_1, x_2 \in X).$$

Take $A = \text{upper cone} \cong \text{cone on } X$, $B = \text{lower cone} = \text{“slightly extended cone”}$; then $A \cup B = X$ and $\tilde{H}_n(A) \cong \tilde{H}_n(B) \cong 0$ since any cone on X is contractible. So by Mayer-Vietoris:

$$\tilde{H}_n(\Sigma X) \cong \tilde{H}_n(A \cap B) \cong \tilde{H}_{n-1}(X) \quad \text{by definition / reduction.}$$

Definition 11.11. Degree of a map $f : S^n \rightarrow S^n$ is $f_*(1) \in \mathbb{Z}$ where $f_* : H_n(S^n) \rightarrow H_n(S^n)$ and $1 \in \mathbb{Z} \cong H_n(S^n)$.

Example 11.3.

- (1) $(\text{Id})_* = \text{id} \implies \deg \text{Id} = 1$
- (2) If $f : S^n \rightarrow S^n$ is not a surjection then $\deg f = 0$ (potential final problem). **TODO:** For f we can assume that f misses a pt $\in S^n$ so we can factor f as a map $S^n \xrightarrow{\bar{f}} S^n \setminus \{x_0\} \hookrightarrow S^n$ so $f_* = (i \circ \bar{f})_* = (i_*)(\bar{f})_*$ so $\deg = 0$
- (3) If $f \simeq g$ then $\deg f = \deg g$.
- (4) $\deg(f \circ g) = \deg f \times \deg g$
- (5) If f is a homotopy equivalence then $\deg f = \pm 1$.
- (6) If f is a reflection along a equitorial sphere then $\deg f = -1$. (Hatcher, Chapter 2.1)
- (7) If f is the anti-podal map on S^n then $\deg f = (-1)^{n+1}$.
- (8) If $f : S^n \rightarrow S^n$ has no fixed points then $\deg f = (-1)^{n+1}$.

Proof. If $f(x) \neq x$ for all $x \in S^n$ then the line segment joining $-x$ to $f(x)$ does not pass through the origin. Then,

$$f_t(x) = \frac{(1-t)f(x) + t(-x)}{|(1-t)f(x) + t(-x)|}$$

defines a homotopy of f to the antipodal map in S^n . So, $f_t : S^n \rightarrow S^n$. □

Theorem 11.12. S^n has a non-zero vector field if and only if n is odd.

Proof. ($\implies$)

Vector field in S^n corresponds to $S^n \xrightarrow{v} \mathbb{R}^{n+1}$, $x \mapsto v(x)$ such that $x \perp v(x) \iff \langle x, v(x) \rangle = 0$. Suppose v is such a vector field on S^n . We may normalize $v(x)$ by $\frac{v(x)}{|v(x)|}$ so that v can be thought of as a map $S^n \xrightarrow{v} S^n$. Now consider $\cos(t) \cdot x + \sin(t) \cdot v(x)$ By letting t go from 0 to π , we obtain a homotopy $f_t : S^n \rightarrow S^n$ from the identity to the antipodal map.

$$\implies \deg \text{id} = \deg(-\text{id}) \implies 1 = (-1)^{n+1} \implies n \text{ is odd.}$$

($\impliedby$)

Suppose n is odd. Say $n = 2k - 1$. Define $v(x_1, \dots, x_{2k}) = (-x_2, x_1, \dots, -x_{2k}, x_{2k-1}) \in \mathbb{R}^{n+1}$.

$x \perp v(x)$ and $\|v(x)\| = 1$. □

Definition 11.13. Euler characteristic of a space X

$$\chi(X) := \sum_n (-1)^n \text{rank}(H_n(X))$$

rank of a finitely-generated, free, abelian group is the number of $\mathbb{Z}$ -summands in that group.

Example 11.4. $\text{rank}(\mathbb{Z} \otimes \mathbb{Z} \otimes \mathbb{Z}) = 1$. $\chi(T^2) = (-1)^0(1) + (-1)^1(2) + (-1)^2(1)$